Market-Level Tug-of-War and Asset Pricing *

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Abstract

We show that the empirical failure of the Capital Asset Pricing Model (CAPM) is substantially less pervasive than previously documented. We construct a simple market-level measure of the tug-of-war between overnight and intraday traders and classify trading days as quiet or noisy. The CAPM performs well on quiet days, which account for half of all trading days, but fails on noisy days. These findings are distinct from previously documented conditional failures of the CAPM and persist after excluding major information events. We provide evidence consistent with over-correction by arbitrageurs as a mechanism underlying the CAPM failure on noisy days.

JEL classification: G11, G12, G41

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I Introduction

The Capital Asset Pricing Model (CAPM) (Sharpe, 1964; Lintner, 1965; Mossin, 1966) is a foundational framework in financial economics, yet its empirical relevance remains the subject of intense scrutiny. A large literature documents the model's failure to explain the cross-section of expected excess returns, with estimated betas often bearing little relation to expected excess returns and, in some cases, the beta–return relation even exhibiting the wrong sign (Fama and French, 1992; Frazzini and Pedersen, 2014). At the same time, the CAPM continues to play a central role in practice: both corporate managers and investors appear to rely on it to guide capital allocation decisions (Graham and Harvey, 2001; Berk and Van Binsbergen, 2016). Recent work attempts to reconcile this apparent disconnect by showing that the CAPM performs well on some high information days, such as those with macroeconomic news (Savor and Wilson, 2014) or earnings announcements (Chan and Marsh, 2022).

We show that the CAPM performs well on a broader set of days than previously documented. In so doing, we make three main contributions. First, we propose a simple, ex-post, and bottom-up measure of aggregate tug-of-war to identify noisy (n) and quiet (q) days. For each stock and each trading day, we define a tug-of-war episode as a positive overnight return, defined as in Lou et al. (2019), followed by a negative intraday return (Akbas et al., 2022). We then compute the aggregate tug-of-war as the fraction of stocks experiencing such negative intraday reversals on a given day. We classify days with aggregate tug-of-war greater than the sample median value of 28% as n-days and the remaining days as q-days. We use daily data on U.S. stocks from July 1992 through December 2022 to implement our

methodology. We find that the average daily market excess return is positive on q-days (0.12) and negative on n-days (-0.05).[1]

Second, we provide a systematic test of the CAPM across these two types of days. Figure 1 illustrates our results, revealing a stark contrast between n- and q-days: the security market line (SML) slopes downward on n-days but upward on q-days.[2] We formally test the model using decile (value-weighted) beta-sorted portfolios and the Fama and MacBeth (1973) methodology. Specifically, for each month and test asset, we use a trailing window of 252 daily excess returns to estimate the conditional beta. For each day in the following month, we regress the cross-section of excess returns on a constant and the conditional beta estimates and obtain daily intercept (α) and slope (λ) estimates. We then average the estimates across all days of interest, i.e., n- or q-days in the sample. We find a significantly positive SML on q-days ($\bar{\lambda}^{(q-day)} = 0.15$, t-stat $= 5.15$) and a significantly negative SML on n-days ($\bar{\lambda}^{(n-day)} = -0.13$, t-stat $= -4.23$).[3]

Although several studies of the security market line, including Hong and Sraer (2016), Hendershott et al. (2020), and Chan and Marsh (2022), focus on the sign of the SML slope, the sign of the estimated slope alone does not establish whether the CAPM holds (Pettengill et al., 1995; Ernst et al., 2021). The CAPM implies that the SML slope equals the average realized market excess return. In a finite-sample regression, however, the average realized market excess return can be either positive or negative. Thus, a negative slope estimate

[1] Throughout the paper, we express all returns in percentage points.

[2] The asset menu consists of 10 (value-weighted) beta-sorted portfolios. The vertical axis shows the average excess returns of each asset for the subsamples of n- or q-days, represented by the blue and red colors, respectively. At the end of each month, we use a trailing window of 252 days to estimate the conditional beta of each portfolio. We then calculate the sample mean of the conditional beta estimates and report it on the horizontal axis.

[3] Throughout the paper, we assess statistical significance at the 5% level.

does not, in itself, constitute evidence against the CAPM. A regression-based test of the CAPM instead involves two restrictions: (i) the average slope equals the average realized market excess return, and (ii) the average intercept equals zero. We therefore focus on these restrictions rather than on the sign of the estimated slope. Our key finding is that the data do not reject the slope restriction on q-days ($\bar{\lambda}^{(q\text{-day})} - (\bar{R}_{m,t}^{(q\text{-day})} - \bar{R}_{f,t}^{(q\text{-day})}) = 0.02$, t-stat $= 1.22$), but strongly reject it on n-days ($\bar{\lambda}^{(n\text{-day})} - (\bar{R}_{m,t}^{(n\text{-day})} - \bar{R}_{f,t}^{(n\text{-day})}) = -0.08$, t-stat $= -3.60$). Moreover, we reject the null hypothesis of an insignificant average intercept on n-days ($\bar{\alpha}^{(n\text{-day})} = 0.06, t\text{-stat} = 3.15$) but not on q-days ($\bar{\alpha}^{(q\text{-day})} = -0.00, t\text{-stat} = -0.25$). Finally, we jointly test the CAPM restrictions for each day type by examining whether the estimated market risk premium equals the corresponding average realized market excess return and whether the mean regression intercept is zero. We do not reject the joint null hypothesis on q-days (p-value $= 0.41$), but reject it on n-days (p-value $= 0.00$).

Third, we provide evidence of a distinct mechanism underlying the failure of the CAPM on n-days, which differs from the mechanisms documented in prior studies. We build on the over-correction hypothesis proposed by Akbas et al. (2022). To fix ideas, suppose that each trading day consists of two trading periods: an overnight period and an intraday period. The overnight period is characterized by high short-selling fees (Bogousslavsky, 2021) and is typically dominated by noise traders, who face leverage constraints and tend to trade around the market open (Berkman et al., 2012). In contrast, institutional investors tend to trade during the rest of the day, making them the dominant force behind intraday returns (Lou et al., 2019, 2024).[4]

When arbitrageurs observe a positive overnight return, they assess the extent to which it

[4]Lou et al. (2019) and Lou et al. (2024) document that household investors are the primary drivers of overnight returns, whereas institutional investors dominate intraday returns.

may be driven by noise-trader demand or fundamental news. If they attribute the return to noise trading, they trade against it during the day, generating a negative intraday return and a tug-of-war for the stock. However, when many stocks simultaneously experience positive overnight returns, arbitrageurs may become overly confident that these returns are noise-driven and underestimate the possibility that some reflect fundamental information. This leads to excessive correction. Since noise traders are constrained by leverage, they are very active in high-beta stocks, resulting in a more pronounced over-correction effect for high-beta stocks. The mechanism predicts that CAPM breakdowns on n-days should be present in the intraday period, stronger when investor sentiment is elevated, and concentrated among non-news days. We find support for each of these predictions.

We examine a range of alternative mechanisms and find that none can account for our results. Savor and Wilson (2014) document that the CAPM does not hold on days without major macroeconomic announcements, including federal funds rate, producer price index, and unemployment rate announcements, while Chan and Marsh (2022) report similar evidence for days without leading earnings announcements (LEADs). Our findings do not simply restate these results. Indeed, n- and q-days occur with similar frequency on macroeconomic announcement days and other days, suggesting that our classification captures a distinct phenomenon. Moreover, restricting the analysis to non-macroeconomic announcement days leaves our conclusions unchanged. The slope restriction continues to hold on q-days ($\bar{\lambda}^{(q\text{-day})} - (\bar{R}_{m,t}^{(q\text{-day})} - \bar{R}_{f,t}^{(q\text{-day})}) = 0.01$, t-stat $= 0.64$), but is rejected on n-days ($\bar{\lambda}^{(n\text{-day})} - (\bar{R}_{m,t}^{(n\text{-day})} - \bar{R}_{f,t}^{(n\text{-day})}) = -0.12$, t-stat $= -5.20$). We obtain similar results when focusing on days that are not classified as LEADs.

We also distinguish our findings from those of Hendershott et al. (2020), who show that

the CAPM holds overnight but fails during the trading hours. One might wonder if our results reflect the greater importance of intraday returns on n-days. To examine this possibility, we separately test the CAPM using overnight and intraday returns on n-days. We reject the slope restriction during both the overnight period ($\bar{\lambda}^{(n\text{-day})} - (\bar{R}_m^{(n\text{-day})} - \bar{R}_f^{(n\text{-day})}) = 0.06$, t-stat $= 5.26$) and the intraday period ($\bar{\lambda}^{(n\text{-day})} - (\bar{R}_m^{(n\text{-day})} - \bar{R}_f^{(n\text{-day})}) = -0.18$, t-stat $= -9.06$). Thus, unlike Hendershott et al. (2020), we find that the CAPM fails during both trading periods on n-days. The two failures, however, may arise from different forces: the overnight failure likely reflects price pressure from noise traders, whereas the intraday failure is consistent with arbitrageurs overcorrecting during the trading hours.

Additionally, we show that the speculative beta mechanism of Hong and Sraer (2016) cannot explain our findings. This mechanism predicts that high-beta assets become susceptible to speculative overpricing, with beta-related distortions and departures from the CAPM becoming more pronounced as disagreement increases. To assess this explanation, we identify periods of high aggregate disagreement following Hong and Sraer (2016) and repeat our asset pricing tests separately for q- and n-days within these periods. We continue to find a significantly positive SML on q-days and a negative one on n-days. More importantly, we strongly reject the slope restriction on n-days during high disagreement periods ($\bar{\lambda}^{(n\text{-day})} - (\bar{R}_{m,t}^{(n\text{-day})} - \bar{R}_{f,t}^{(n\text{-day})}) = -0.09$, t-stat $= -4.04$), while we do not reject it on q-days ($\bar{\lambda}^{(q\text{-day})} - (\bar{R}_{m,t}^{(q\text{-day})} - \bar{R}_{f,t}^{(q\text{-day})}) = -0.01$, t-stat $= -0.47$). Moreover, the rejection is stronger on n-days during periods of low disagreement ($\bar{\lambda}^{(n\text{-day})} - (\bar{R}_{m,t}^{(n\text{-day})} - \bar{R}_{f,t}^{(n\text{-day})}) = -0.15$, t-stat $= -4.12$) than during periods of high disagreement. This pattern is inconsistent with the speculative beta hypothesis, which predicts that disagreement-induced beta mispricing increases with investor disagreement.

Finally, we examine whether the disagreement-based beta compression mechanism of Andrei et al. (2023) can account for our findings. Their mechanism implies that investor disagreement biases beta estimates and, in turn, rotates the SML. In our data, however, conditional beta estimates are remarkably similar on q- and n-days, and we find no evidence of beta compression on q-days. These findings provide little support for beta distortion as an explanation for the differential performance of the CAPM across the two day types.

We conduct a range of additional checks. First, we obtain similar results using pooled regressions. Second, we use equal-weighted, rather than value-weighted, beta-sorted portfolios as test assets; the main findings remain unchanged. Third, we replicate our analysis at the individual stock level, again finding qualitatively similar patterns. Fourth, we expand the test asset universe to address the critique of Lewellen et al. (2010). Specifically, we augment the 10 beta-sorted portfolios with 25 size- and book-to-market-sorted portfolios and 10 Fama–French industry portfolios. Our key conclusions hold across this broader asset menu. Fifth, we test the sensitivity of our results to alternative definitions of n-days. Our main finding remains robust across a wide range of thresholds used to identify n-days. Sixth, we examine the potential impact of microstructure noise and find no evidence to suggest that it materially affects our conclusions. Seventh, we conduct a bootstrap analysis and find that the estimated market risk premium remains negative when we restrict the sample to n-days with positive average realized market excess returns. Taken together, these extensive checks support the conclusion that the different results associated with the slope restriction of the CAPM across n- and q-days are not an artifact of sample construction, measurement error, or methodological choices.

Our paper contributes to the empirical literature on the CAPM. This literature dates

back to Fama and French (1992) who document that the SML is empirically flat. Subsequent research documents time variation in the slope of the SML. Bernanke and Kuttner (2005), Savor and Wilson (2014), and Lucca and Moench (2015) find a positive SML only on macroeconomic announcement days. Antoniou et al. (2016) document an upward sloping SML during periods of pessimistic sentiment and a downward sloping SML during optimistic periods. Hong and Sraer (2016) show how aggregate disagreement affects the sign of the slope of the SML. Jylhä (2018) establishes a negative relationship between the slope of the SML and the prevailing margin requirement. Chan and Marsh (2022) and Cui and Li (2024) document a positive SML on leading earnings announcement days and balanced trading days, respectively. Abolghasemi et al. (2024) relate the slope of the SML to changes in the granularity of the economy. Several studies attempt to rationalize the time variation in the slope of the SML using arguments based on the sign of the average realized market excess return (Ernst et al., 2021), the information gap between investors and the econometrician (Andrei et al., 2023), and small-sample problems (Ghaderi and Seo, 2026). In contrast to prior studies, which focus primarily on the sign of the SML slope and find evidence consistent with the CAPM only over a limited number of trading days, our analysis focuses on the CAPM-implied slope restriction. Our novel finding is that the CAPM holds over a substantially larger sample comprising several thousand q-days. These days span both (i) periods of optimistic and pessimistic investor sentiment and (ii) days with and without major news announcements.

Our work adds to the growing asset pricing literature that exploits intraday and overnight information. Lou et al. (2019) decompose the close-to-close returns into overnight and intraday returns and document a tug-of-war between the overnight and intraday traders. Hender-

shott et al. (2020) compare the performance of the CAPM during the overnight and intraday trading periods. Bogousslavsky (2021) analyzes the cross-section of intraday and overnight expected returns. Akbas et al. (2022) examine the impact of the firm-level tug-of-war on the cross-section of next-month excess returns. Our research brings together the literature on (i) the tug-of-war and (ii) the time-varying slope of the SML. To the best of our knowledge, we are the first to propose a simple indicator variable of the tug-of-war at the market level that identifies n- and q-days and documents the implications for the performance of the CAPM at the daily horizon.

The remainder of the paper proceeds as follows. Section II presents our data and methodology. Section III documents our main results and performs various robustness checks. Section IV discusses the mechanism underlying our main results. Section V evaluates alternative explanations and shows that they do not explain our findings. Finally, Section VI concludes the paper.

II Data and Methodology

A Data

We obtain the stock data, including the opening and closing prices of each trading day, from the Center for Research in Security Prices (CRSP). Since CRSP only reports the opening price from 1992 (Akbas et al., 2022), our sample period starts in July 1992 and ends in December 2022. Our dataset consists of all the U.S.-based common stocks (with a CRSP share code value equal to 10 or 11) traded on the American Stock Exchange (AMEX), the National Association of Securities Dealers Automated Quotations (NASDAQ), and the

New York Stock Exchange (NYSE). We discard any observation with a missing opening or closing price, a zero intraday or overnight return.[5]

We calculate the overnight return of each stock as follows (Lou et al., 2019):

$$R_{i,t,night} = \frac{1 + R_{i,t}}{1 + R_{i,t,day}} - 1 \tag{1}$$

where $R_{i,t,night}$ is the overnight return associated with firm i on day t. $R_{i,t}$ is the close-to-close return of firm i on day t. We obtain this return directly from the CRSP database. Finally, $R_{i,t,day}$ is the intraday, i.e., open-to-close, return of firm i on day t. We compute it as follows: $R_{i,t,day} = \frac{P_{i,t,close}}{P_{i,t,open}} - 1$, where $P_{i,t,close}$ and $P_{i,t,open}$ denote the closing and opening prices of stock i, on day t, respectively.

B Methodology

Tug-of-War Our identification of noisy and quiet days involves three steps. First, we build on the insights of Lou et al. (2019) and Akbas et al. (2022), who identify a tug-of-war for a given stock i on day t when the stock has a positive overnight return followed by a negative

[5]We apply this filter in both the computation of the tug-of-war measure and the construction of the test assets. Untabulated results demonstrate that our main conclusions are quantitatively robust to applying the filter exclusively in computing the tug-of-war measure.

intraday return:[6]

$$ToW_t^{(i)} = \begin{cases} 1 & \text{if } R_{i,t,night} > 0 \ \& \ R_{i,t,day} < 0 \\ 0 & \text{Otherwise} \end{cases} \tag{2}$$

where $ToW_t^{(i)}$ is a dummy variable that takes the value 1 if there is a tug-of-war associated with stock i on day t.

Second, we compute the market-level tug-of-war by averaging the firm-level tug-of-war indicators:[7]

$$ToW_t^m = \frac{\sum_{i=1}^{N_t} ToW_t^{(i)}}{N_t} \tag{3}$$

where ToW_t^m denotes the tug-of-war measure at time t aggregated at the market level. N_t is the total number of stocks observed on day t.

Third, we classify a day as noisy (n) if the ToW^m observed on that day exceeds the full sample median estimate of 0.28 and as quiet (q) otherwise.[8] Figure 2 shows the share of n-days over time. We can see that in the earlier part of the sample period, the fraction of n-days was typically higher than 50%, whereas it is lower in the more recent sample, with

[6]One might ask: why do Akbas et al. (2022) focus on the positive overnight return followed by an intraday reversal and not the opposite, i.e., a negative overnight return followed by a positive intraday return? This choice is motivated by several findings in the literature. First, retail traders are mostly active around the market open (Berkman et al., 2012), while institutional investors trade mostly during the rest of the day (Lou et al., 2019, 2024). Second, the high overnight short selling fees/constraints (Bogousslavsky, 2021) make it difficult for retail traders to open short positions. In turn, this observation implies that retail investors are unlikely to open short positions if they have a negative view of the company. Section IV.D presents empirical results consistent with this intuition.

[7]We also implement a "direct approach" by computing the tug-of-war directly on the value-weighted CRSP index, and find that the main result is very similar. See Section III.C.4 for further details.

[8]One might ask to what extent the results depend on the use of the median estimate instead of the average estimate. Our untabulated analysis shows that there is very little to distinguish between these two choices. In the data, the median ToW^m computed over the sample is equal to 0.28. This estimate is the same as the mean estimate (see Panel A of Table 1). Section III.C.3 discusses alternative cut-offs, including that of 0.25, and shows that the results are robust.

the minimum value around 40%. Panel A of Table 1 shows the key moments of the aggregate tug-of-war time-series. We can see that the mean and median estimates are the same. The table also shows that there is very little persistence in the series. Panels B and C of the same table show that both types of days are equally likely to occur on a given (i) day of the week or (ii) month of the year. Given this finding, we conclude that the seasonal effects documented in Tinic and West (1984) and Birru (2018) do not drive our main results. We also compare the values of key state variables on n- and q-days. Specifically, we look at the (i) TED spread, (ii) VIX, and (iii) risk aversion index (Bekaert et al., 2022).[9] Panel D of Table 1 shows that there is generally very little to distinguish between the two types of days. The only notable exception is the average realized market excess return, which is negative on n-days (-0.05) and positive on q-days (0.12), respectively. Given these estimates, we not only analyze the sign of the security market line slope but also formally test the slope restriction for each day type.

Test Assets Our main objective is to assess the performance of the CAPM on n- and q-days, respectively. To achieve this goal, we use 10 beta-sorted portfolios as test assets. We construct these portfolios as follows. For each security and month-end date, we use a trailing window of 252 daily close-to-close returns to estimate the (pre-ranking) beta of the

[9]We obtain our time-series of the TED spread from the website of the Federal Reserve Bank of St Louis. This dataset ends on 28 January 2022. The VIX data come from Bloomberg. The risk aversion index is available on the following website: https://www.nancyxu.net/risk-aversion-index.

security:[10]

$$R_{i,t} - R_{f,t} = \alpha_i + \beta_i(R_{m,t} - R_{f,t}) + \epsilon_{i,t} \tag{4}$$

where $R_{f,t}$ denotes the risk-free rate, proxied by the 1-month Treasury bill rate, observed on day t. α_i and β_i are the intercept and slope parameters associated with security i, respectively. $R_{m,t}$ is the return of the market portfolio at time t, proxied by the value-weighted CRSP index. The data for both the risk-free rate and the market portfolio come from Kenneth French's website.[11] $\epsilon_{i,t}$ is the residual associated with stock i at time t.

We then use the (pre-ranking) beta estimates to sort the stocks into 10 beta portfolios to hold for the next month. This allows us to calculate the daily value-weighted returns of each beta portfolio over the month.[12] By repeating the above steps for each calendar month, we obtain the daily time-series of returns for each decile portfolio, where individual portfolios are rebalanced on a monthly basis.

III Main Results

This section presents our main results. We begin with the Fama and MacBeth (1973) analysis. Next, we implement a pooled estimation. Finally, we perform several robustness checks and show that they do not alter our main findings.

[10] Our interest in this trailing window is motivated by the work of Hollstein et al. (2019), which documents that this estimation window performs well empirically. Throughout this paper, we require the availability of at least 126 observations in order to use the trailing window.

[11] The website is available at the following address: http://mba.tuck.dartmouth.edu/pages/faculty/ken.french.

[12] As a robustness check, we also consider an equal-weighting scheme. This alternative does not materially affect our main conclusions. See Section III.C.1 for further details.

A Fama and MacBeth (1973) Regression

All Days We implement the empirical procedure of Fama and MacBeth (1973) as follows. For each test asset i and month m, we estimate the conditional (post-ranking) beta, denoted $\hat{\beta}_{i,m}$, using the most recent 252 daily observations (see Equation (4)). Subsequently, for each trading day t in the following month $m+1$, we run the cross-sectional regression of daily portfolio excess returns on a constant and the conditional portfolio betas estimated at the end of month m:

$$R_{i,t} - R_{f,t} = \alpha_t + \lambda_t \hat{\beta}_{i,m} + \epsilon_{i,t}, \tag{5}$$

where $R_{i,t} - R_{f,t}$ denotes the daily excess return of asset i on day t of month $m+1$. The parameters α_t and λ_t represent the intercept and the market risk premium estimates for day t, respectively, while $\epsilon_{i,t}$ is the residual term.

Repeating this cross-sectional regression for each day yields a daily time series of parameter estimates (α_t and λ_t) within month $m+1$. We then iterate this procedure across all months in the sample to construct the full time series of estimated parameters, which we use for statistical inference. Throughout this paper, we compute the standard errors following Newey and West (1987), with the bandwidth of the Bartlett kernel set to 10.[13]

We test the following predictions of the CAPM. First, the average slope estimate is significantly different from zero. Second, the average slope estimate is equal to the average realized excess return of the market. Throughout this paper, we refer to this restriction as the slope restriction. The first set of results in Table A1 of the Online Appendix shows

[13]This choice is based on the observation-based criterion of the Newey and West (1987) estimator: $4(T/100)^{2/9}$, where T is the total number of observations. We round the estimate to the nearest integer.

that, when we use information from all days in our sample, the relationship between beta and excess return is flat, as evidenced by the insignificant average slope estimate ($\bar{\lambda} = 0.01$, t-stat=0.56).[14] Furthermore, the average intercept is significantly different from zero ($\bar{\alpha} = 0.03$, t-stat=2.28). Collectively, these results are difficult to reconcile with the key predictions of the CAPM.

n- versus q-Days Next, we examine the empirical results obtained when restricting the analysis to n-days and q-days separately. We employ a two-step procedure. First, we compute the conditional (post-ranking) betas ($\hat{\beta}_{i,m}$) as described previously. Second, for each month $m+1$, we estimate the cross-sectional regression using only the daily returns corresponding to the day-type of interest—either n- or q-days—as the dependent variable. Specifically, we estimate:

$$R_{i,t}^{(n\text{-day})} - R_{f,t}^{(n\text{-day})} = \alpha_t^{(n\text{-day})} + \lambda_t^{(n\text{-day})} \hat{\beta}_{i,m} + \epsilon_{i,t}^{(n\text{-day})}, \tag{6}$$

$$R_{i,t}^{(q\text{-day})} - R_{f,t}^{(q\text{-day})} = \alpha_t^{(q\text{-day})} + \lambda_t^{(q\text{-day})} \hat{\beta}_{i,m} + \epsilon_{i,t}^{(q\text{-day})}, \tag{7}$$

where α_t and λ_t denote the intercept and the estimated market risk premium on day t of month $m+1$, respectively. The superscripts n-day and q-day indicate that the estimation is performed using observations from days classified as n- and q-days, respectively.[15]

Starting with n-days, Panel A of Table 2 shows that the time-series average of the in-

[14] We obtain similar findings across alternative choices of test assets, e.g., individual equities (see Table A1 of the Online Appendix).

[15] Since the analysis uses each portfolio's conditional beta, estimated at the end of the preceding month using all daily returns, including both n- and q-days, as the explanatory variable, an alternative approach would be to estimate post-ranking betas separately using only n-day or q-day observations. Figure 3 plots the betas estimated using q-days against those estimated using n-days and shows that the two sets of estimates are highly similar.

tercept is positive and highly significant on n-days ($\bar{\alpha}^{(n-day)} = 0.06$, t-stat=3.15). Moreover, the estimated market risk premium is significantly negative ($\bar{\lambda}^{(n-day)} = -0.13$, t-stat=-4.23). The last column establishes that the estimated market risk premium differs significantly from the average realized market excess return computed over n-days ($\bar{\lambda}^{(n-day)} - (\bar{R}_{m,t}^{(n-day)} - \bar{R}_{f,t}^{(n-day)}) = -0.08$, t-stat=-3.60).

Turning to q-days, we see that the average intercept is small and statistically indistinguishable from 0 ($\bar{\alpha}^{(q-day)} = -0.00$, t-stat=-0.25). Furthermore, the estimated market risk premium is positive and statistically significant ($\bar{\lambda}^{(q-day)} = 0.15$, t-stat=5.15). Consistent with the CAPM, this estimated risk premium is statistically indistinguishable from the average realized market excess return computed over q-days ($\bar{\lambda}^{(q-day)} - (\bar{R}_{m,t}^{(q-day)} - \bar{R}_{f,t}^{(q-day)}) = 0.02$, t-stat=1.22).

We go a step further and test—separately for each day-type—the joint restriction that the estimated market risk premium equals the average market excess return observed on days of that type, and that the average regression intercept is zero. This joint restriction is not rejected on q-days (p-value = 0.41), but is strongly rejected on n-days (p-value = 0.00), underscoring the contrasting performance of the CAPM across day types.

Overall, the results suggest that the empirical performance of the asset pricing model depends on the type of days analyzed. On n-days, the empirical performance of the CAPM is at odds with its predictions. On q-days, however, the CAPM provides a good description of the empirical data.

B Pooled Regression

All Days We now perform a pooled estimation:

$$R_{i,t} - R_{f,t} = \alpha + \lambda \hat{\beta}_{i,m} + \epsilon_{i,t} \tag{8}$$

where all variables are as previously described. Throughout this paper, our pooled estimation includes day fixed effects and standard errors are clustered by trading day.

Panel B of Table 2 documents a significantly positive intercept ($\alpha = 0.05$, $t-$stat$=3.25$). Furthermore, the relationship between beta and excess returns is not significant ($\lambda = -0.01$, t-stat$=-0.33$). These findings are broadly in line with those of Table A1 of the Online Appendix.

***n*- versus *q*-Days** Next, we consider an extended model where the main explanatory variable is interacted with a dummy variable that takes value 1 if the excess return of the portfolio is observed on an n-day:

$$R_{i,t} - R_{f,t} = \alpha + \lambda \hat{\beta}_{i,m} + \psi \mathbb{1}_t^{n-day} + \phi \hat{\beta}_{i,m} \times \mathbb{1}_t^{n-day} + \epsilon_{i,t} \tag{9}$$

where α and λ are the intercept and market risk premium associated with q-days, respectively. ψ and ϕ denote the differences in the intercept and slope parameters, respectively, between n- and q-days. $\mathbb{1}_t^{n-day}$ is a dummy variable that takes value 1 if the excess return of the test asset is observed on an n-day and 0 otherwise. All other variables are as defined above.

Panel B of Table 2 highlights several results. First, the intercept estimated on q-days ($\alpha = -0.00$, t-stat$=-0.01$) is small and statistically indistinguishable from 0. This result is

consistent with the CAPM. Second, the estimated market price of risk is positive and highly significant ($\lambda = 0.14$, t-stat=4.48). Given these results, we conclude that there is a positive risk-return trade-off on q-days, as predicted by the CAPM. The parameter estimates based on the pooled estimation are statistically indistinguishable from those based on the Fama and MacBeth (1973) methodology (see Panel A of Table 2).

Overall, our results point to the contrasting performance of the CAPM across n- and q-days. The model fails on n-days but performs well on q-days. Consistent with the CAPM, the estimated market risk premium is statistically indistinguishable from the average realized market excess return on q-days, whereas this restriction is rejected on n-days. These findings are robust to both the Fama and MacBeth (1973) methodology and the pooled regression approach.

C Robustness Checks

Having established our main findings, we now proceed to assess their robustness to alternative research design specifications. To begin with, we examine the extent to which the portfolio-level weighting scheme may affect our main results. We then assess the sensitivity of the results to the menu of test assets. Next, we consider different methodological choices for the identification of n- and q-days. Additionally, we assess the impact of microstructure noise on the results. Finally, we conduct a bootstrap exercise to examine the impact of the sign of the average realized market excess return on our findings.

1 The Portfolio Weights

Our main analysis uses value-weighted returns, which limit the influence of microcap stocks. We assess the sensitivity of our results to this choice by repeating the analysis using equal weights. The results are reported in Table A2 of the Online Appendix. Our main findings are similar. In particular, the slope restriction is not rejected on q-days ($\bar{\lambda}^{(q-day)} - (\bar{R}_{m,t}^{(q-day)} - \bar{R}_{f,t}^{(q-day)}) = 0.03$, t-stat=1.65), whereas it is rejected on n-days ($\bar{\lambda}^{(n-day)} - (\bar{R}_{m,t}^{(n-day)} - \bar{R}_{f,t}^{(n-day)}) = -0.20$, t-stat=-9.34).

2 Menu of Assets

Up to this point, our asset pricing tests focus on 10 beta-sorted portfolios. One might wonder about the robustness of the results to the choice of test assets. To shed some light on this, we follow a two-pronged strategy. First, we use individual stocks as test assets. Second, we extend the set of 10 beta-sorted portfolios to 45 portfolios.

Individual Equities Table A3 of the Online Appendix shows the main results when using individual equities as test assets. Panel A focuses on the Fama and MacBeth (1973) results. We observe a significant and positive market risk premium on q-days ($\bar{\lambda}^{(q-day)}$=0.10, t-stat=5.68), which is not significantly different from the average realized market excess return ($\bar{\lambda}^{(q-day)} - (\bar{R}_{m,t}^{(q-day)} - \bar{R}_{f,t}^{(q-day)}) = -0.02$, t-stat=-1.78). In contrast, the slope restriction is rejected on n-days ($\bar{\lambda}^{(n-day)} - (\bar{R}_{m,t}^{(n-day)} - \bar{R}_{f,t}^{(n-day)}) = -0.11$, t-stat=-8.57).

We next control for size ($Size$), book-to-market (BTM), and momentum (MOM). We measure the size of a firm as the logarithm of its market capitalization. Our calculation of BTM follows Fama and French (1992). We compute a firm's MOM of month t as the

cumulative stock return over the last 11 months, skipping the most recent month (Daniel and Moskowitz, 2016). Panel B shows that the inclusion of control variables in the implementation of the Fama and MacBeth (1973) analysis does not materially affect the main results. Panel C of Table A3 of the Online Appendix further presents similar results, in terms of both magnitude and statistical significance, in a pooled regression setting.

A Broader Menu of Assets Mindful of the Lewellen et al. (2010) critique, we expand the menu of test assets in our baseline analysis. Specifically, we augment the 10 (value-weighted) beta-sorted portfolios with (i) 25 size- and book-to-market-sorted portfolios and (ii) 10 Fama–French industry portfolios.

Panel A of Table A4 of the Online Appendix reports coefficient estimates of the Fama and MacBeth (1973) regressions for the 45 test portfolios. The estimated market risk premium is negative on n-days ($\bar{\lambda}^{(n\text{-day})} = -0.17$, t-stat $= -5.71$) and positive on q-days ($\bar{\lambda}^{(q\text{-day})} = 0.15$, t-stat $= 5.06$). The slope restriction is not rejected on q-days ($\bar{\lambda}^{(q-day)} - (\bar{R}_{m,t}^{(q-day)} - \bar{R}_{f,t}^{(q-day)}) = 0.03$, t-stat$=1.38$), whereas it is rejected on n-days ($\bar{\lambda}^{(n-day)} - (\bar{R}_{m,t}^{(n-day)} - \bar{R}_{f,t}^{(n-day)}) = -0.12$, t-stat$=-5.46$). Panel B reports similar estimates based on a pooled estimation.

3 Alternative Thresholds

Our baseline identification of the n- and q-days depends on whether or not the aggregate tug-of-war measure is higher than its sample median. Clearly, this threshold is somewhat arbitrary and may introduce misclassification errors into our analysis. Given this, it is prudent to consider alternative thresholds. We consider two alternative thresholds. First, we use a threshold of 0.25. Our interest in this value is motivated by the intuition that, under

standard assumptions, there is a 25% probability of observing a positive overnight return followed by a negative intraday return for a given day and firm in our cleaned dataset.[16] Second, we set the threshold equal to the 70th percentile of the full-sample distribution of the aggregate tug-of-war. For each of these two cases, we classify a day as n-day if the daily aggregate tug-of-war variable exceeds the specified threshold.

Tables A5 and A6 of the Online Appendix present the results of the Fama and MacBeth (1973) and pooled regression analyses. We do not reject the slope restriction on q-days, whereas we reject it on n-days. We therefore conclude that our main results are robust to potential classification errors.

4 Potential Microstructure Noise

One concern relates to the potential impact of microstructure noise, which may arise from frictions such as the discreteness of price changes and the bid–ask bounce. Theoretically, microstructure noise could affect our baseline analysis in at least two ways. First, it may affect the firm-level estimates of the tug-of-war. Second, it could directly affect the asset pricing tests.

In order to address the first concern, we calculate the market-level tug-of-war by using the returns of the stock market index directly. Specifically, we estimate the market-level tug-of-war as follows:

[16] Recall that we discard observations associated with a missing opening or closing price, a zero intraday or overnight return (see Section II.A).

$$ToW_t^{(CRSP)} = \begin{cases} 1 & \text{if } R_{CRSP,t,night} > 0 \ \& \ R_{CRSP,t,day} < 0 \\ 0 & \text{Otherwise} \end{cases} \tag{10}$$

where $ToW_t^{(CRSP)}$ is a dummy variable that takes the value 1 if there is a tug-of-war associated with the value-weighted CRSP index on day t. $R_{CRSP,t,day}$ and $R_{CRSP,t,night}$ denote the intraday and overnight returns associated with the value-weighted CRSP index on day t, respectively.

Armed with this measure, we identify n- and q-days as days on which the daily estimate is 1 and 0, respectively.[17] We then repeat our main empirical tests. Table A7 of the Online Appendix shows that this methodological change leads to qualitatively similar conclusions. We do not reject the slope restriction on q-days, whereas we reject it on n-days.

Turning to the second concern, Asparouhova et al. (2010) document that the microstructure noise may bias the ordinary least squares (OLS) parameter estimates. Since our analysis focuses primarily on value-weighted portfolios as test assets, this is not a significant concern. However, it is possible that the microstructure noise may bias our estimates based on individual stocks as test assets. To address this concern, we implement the weighted least squares (WLS) estimation recommended by the authors. Specifically, we weigh each observation by 1 plus the lagged daily return of asset i, i.e., $1 + R_{i,t-1}$. Table A8 of the Online Appendix summarizes the findings. Comparing these estimates with those in Table A3 of the Online Appendix, we find that there is very little difference between the two sets of results, suggesting that potential microstructure noise does not materially affect our results.

[17] Empirically, we find that 96.34% of the n-days based on this "direct" aggregate tug-of-war correspond to the n-days based on the bottom-up approach used in our baseline estimation.

5 Bootstrap Exercise

The CAPM's risk–return tradeoff pertains to expected, rather than realized, returns. Following standard practice, we use realized market excess returns as proxies for expected returns, which is reasonable when evaluating the model over a long sample period. However, Pettengill et al. (1995) show that a negative average realized market excess return can generate a negative observed relationship between beta and excess returns. Consequently, the regression-based estimate of the market risk premium may mechanically reflect the sign of the average realized market excess return (Ernst et al., 2021). This concern is relevant to tests based on the sign of the estimated market risk premium, but does not apply to our main analysis, which focuses on the CAPM-implied slope restriction. It nevertheless raises the possibility that the contrasting signs of the estimated market risk premia on n- and q-days partly reflect the differences in average realized market excess returns across the two day types. Indeed, Panel D of Table 1 shows average market excess returns of -0.05 on n-days and 0.12 on q-days.

To ensure our findings are not driven by the sign of the average market excess return, we implement a conditional bootstrap exercise. The aim is to create a pseudo-sample with the same number of n-days as our original sample. We randomly select 3,720 days (with replacement) from the n-day sample. If the average market excess return of this pseudo-sample is negative, we discard it and repeat the procedure until we obtain a pseudo-sample with a positive average market excess return. We then perform the estimation outlined in Fama and MacBeth (1973) and compute the market risk premium estimate. We repeat these steps 1,000 times.

Figure A1 of the Online Appendix shows the histogram of the estimated market risk

premium from the 1,000 pseudo samples. The mean is negative, averaging −0.12, despite the fact that the average realized market excess returns of each pseudo sample is positive. This result suggests that our finding of a negatively sloped SML on n-days is not a mechanical result due to the sign of the average realized market excess return.

IV What Explains the Results?

Our main intuition builds on the over-correction hypothesis proposed by Akbas et al. (2022). For ease of exposition, we decompose each close-to-close trading day into an overnight period and an intraday period. The overnight period is characterized by high short-selling fees (Bogousslavsky, 2021) and is primarily driven by noise traders, who tend to trade near the market open (Berkman et al., 2012). In contrast, institutional investors tend to trade throughout the rest of the day and therefore dominate intraday returns (Lou et al., 2019). Consequently, a positive overnight return may reflect favorable overnight news, such as an earnings announcement released while the market is closed, noise-trader-induced price pressure, or a combination of the two.

During the intraday period, arbitrageurs assess whether the overnight return primarily reflects news or noise. If they conclude that the overnight return is largely driven by noise trading, they trade against the resulting price pressure, generating a negative intraday return and a tug-of-war. Akbas et al. (2022) argue that repeated tug-of-war episodes for the same stock cause arbitrageurs to underestimate the probability that positive overnight returns reflect genuine information. As a result, they place greater weight on the possibility that the initial price increase is driven by noise trading and take larger positions against it, leading to

over-correction. We extend this idea by arguing that arbitrageurs can also use information from the cross-section to infer the likelihood of noise trading. Specifically, when arbitrageurs observe positive overnight returns across many stocks on the same day, e.g., on n-days, they may infer that noise traders are unusually optimistic and become excessively aggressive in correcting prices. As a result, they discount the possibility that some of the overnight returns reflect fundamental news and over-correct during the trading hours.

Because optimistic, leverage-constrained noise traders tend to concentrate their trading in high-beta stocks (Barber and Odean, 2007; Hong and Sraer, 2016), the resulting over-correction should be strongest for these stocks. This prediction is consistent with our empirical evidence (see Figure 1), which shows that (i) the difference in average daily excess returns between q- and n-days increases with beta and (ii) the downward-sloping security market line on n-days is driven primarily by high-beta assets. Overall, the over-correction mechanism predicts a negatively sloped SML and explains why the slope restriction is rejected on n-days. In the remainder of this section, we present and test several additional implications of this mechanism.

A Dissecting the SML on n-days

The over-correction hypothesis suggests that the overcorrection occurs during the intraday trading period, when arbitrageurs are active. A natural implication is that the rejection of the slope restriction on n-days should be pronounced when analyzing the intraday period. To examine this prediction, we focus on n-days and estimate the cross-sectional asset pricing

model separately using (i) overnight and (ii) intraday excess returns.[18]

Panel A of Table 3 reports a positive and significant SML during the overnight period ($\bar{\lambda}^{(n\text{-day})} = 0.36$, t-stat $= 19.10$) and a negative and significant SML during the intraday period ($\bar{\lambda}^{(n\text{-day})} = -0.55$, t-stat $= -20.45$). This pattern is consistent with the evidence reported by Hendershott et al. (2020). More importantly, we reject the slope restriction during both trading periods. During the overnight period, we find $\bar{\lambda}^{(n\text{-day})} - (\bar{R}_{m,t}^{(n\text{-day})} - \bar{R}_{f,t}^{(n\text{-day})}) = 0.06$ (t-stat $= 5.28$), suggesting that prices already deviate from CAPM predictions before arbitrageurs begin trading. During the intraday period, the rejection becomes substantially stronger, with $\bar{\lambda}^{(n\text{-day})} - (\bar{R}_{m,t}^{(n\text{-day})} - \bar{R}_{f,t}^{(n\text{-day})}) = -0.18$ (t-stat $= -9.06$). Overall, the evidence is consistent with the notion that noise traders induce mispricing overnight, which is subsequently over-corrected by arbitrageurs during the trading day.

B News Days

The over-correction hypothesis depends crucially on whether arbitrageurs attribute positive overnight returns to news or noise trading. If arbitrageurs believe that the news component dominates, they are less likely to over-correct.[19] Since our tug-of-war measure is constructed at the market level, we focus on news events that are likely to contain information about the broader economy. We first analyze macroeconomic news days and then turn our attention to LEAD days.

[18] Following Lou et al. (2024), we compute overnight and intraday excess returns by subtracting the corresponding fraction of the daily risk-free rate from raw returns, namely 17.5/24 for the overnight period and 6.5/24 for the intraday trading window.

[19] In an untabulated analysis, we confirm the findings of Bernanke and Kuttner (2005), Savor and Wilson (2014), and Chan and Marsh (2022) that the SML is positively sloped on news days.

Macroeconomic News Days Building on the work of Savor and Wilson (2014), we split our sample of trading days into news days and other days. News days are associated with announcements of the (i) producer price index, (ii) unemployment rate, and (iii) federal funds rate.[20] We refer to all remaining days as other days. For each subsample, we repeat our main asset pricing tests. Panel A of Table 4 focuses on the results for n-days, for which we investigate whether the failure of the CAPM is driven by macroeconomic news or other days.

Several results emerge. To begin with, the SML is flat on n-days that overlap with macroeconomic news days ($\bar{\lambda}^{(n\text{-day})} = -0.08$, t-stat $= -0.98$), likely reflecting the insignificant average realized market excess return on these days. In contrast, the SML is negatively sloped on n-days that overlap with other days ($\bar{\lambda}^{(n\text{-day})} = -0.18$, t-stat $= -5.74$). Moreover, we strongly reject the slope restriction on other days: $\bar{\lambda}^{(n\text{-day})} - (\bar{R}_{m,t}^{(n\text{-day})} - \bar{R}_{f,t}^{(n\text{-day})}) = -0.12$ (t-stat $= -5.20$). This pattern is consistent with the over-correction hypothesis, which predicts that arbitrageurs are more likely to over-correct when positive overnight returns are unlikely to reflect broad-based information.

Leading Earnings Announcement Days Following Chan and Marsh (2022), we identify LEAD days as those falling between Tuesday and Thursday of the first week of the earnings reporting quarter, provided that at least 50 S&P 500 firms announce earnings during that week. All other days are classified as non-LEAD days.[21] Since several important firms announce earnings on LEAD days, arbitrageurs are more likely to attribute positive overnight

[20] Savor and Wilson (2014) explain that, for the period after 1972, the announcement of the producer price index typically takes place a few days before that of the consumer price index. Since our sample period starts in 1992, we therefore focus on the producer price index.

[21] Panel D of Table 1 shows that n days are equally likely to overlap with LEADs and other days.

returns on these days to fundamental news rather than noise trading. Therefore, the over-correction hypothesis predicts stronger evidence of over-correction on n-days associated with non-LEAD days than on n-days associated with LEAD days.

The results in Panel A of Table 5 support this prediction. The estimated market risk premium is insignificant on n-days that overlap with LEADs ($\bar{\lambda}^{(n\text{-day})} = 0.12$, t-stat $= 0.69$). Moreover, we do not reject the slope restriction (t-stat $= 1.42$) on these days. Taken together, these results suggest that the flat SML observed on these days does not indicate a rejection of the CAPM; rather, it is consistent with the average realized market excess return being statistically indistinguishable from zero. In contrast, on n-days that overlap with other days, we observe a significantly negative SML ($\bar{\lambda}^{(n\text{-day})} = -0.15$, t-stat $= -3.77$). Moreover, the slope restriction is rejected on non-LEAD days ($\bar{\lambda}^{(n\text{-day})} - (\bar{R}_{m,t}^{(n\text{-day})} - \bar{R}_{f,t}^{(n\text{-day})}) = -0.12$, t-stat $= -4.14$). This pattern suggests that arbitrageurs are more likely to over-correct when positive overnight returns are less likely to reflect the arrival of fundamental information.

C The Impact of Sentiment

Another testable prediction of the over-correction hypothesis is that arbitrageurs are more likely to over-correct when noise trading activity is expected to be higher. To test this prediction, we build on the investor sentiment index of Baker and Wurgler (2006).[22] Specifically, following Antoniou et al. (2016), we classify trading months as optimistic or pessimistic based on the lagged investor sentiment index. A month is classified as optimistic (pessimistic) if the sentiment score measured at the end of the previous month is positive

[22]We obtain the monthly index from https://pages.stern.nyu.edu/~jwurgler/data/Investor_Sentiment_Data_20190327_POST.xlsx.

(negative). For each sentiment period, we repeat our main asset pricing tests focusing on n-days.

Several findings emerge from Panel A of Table 6. First, the SML is negatively sloped on n-days during both pessimistic ($\bar{\lambda}^{(n\text{-day})} = -0.13$, t-stat $= -3.02$) and optimistic ($\bar{\lambda}^{(n\text{-day})} = -0.26$, t-stat $= -7.84$) periods. The stronger downward slope during optimistic periods is consistent with the over-correction hypothesis, as arbitrageurs are more likely to attribute positive overnight returns to noise trading when sentiment is elevated. Second, we reject the slope restriction during both sentiment regimes.

D Positive Intraday Reversal

The costs associated with holding short positions overnight (Bogousslavsky, 2021) constrain noise traders' ability to initiate such positions. Consequently, arbitrageurs are likely to over-correct primarily following positive overnight returns, because the initial price movement is in the direction in which noise traders can more readily establish positions. This mechanism therefore yields another prediction: a reversal characterized by a negative overnight return followed by a positive intraday return should not be associated with over-correction. If anything, the CAPM should hold on such days.

To examine this prediction, we construct an alternative measure of firm-level tug-of-war that identifies these positive intraday reversals:

$$ToW_t^{\text{alternative},(i)} = \begin{cases} 1 & \text{if } R_{i,t,\text{night}} < 0 \ \& \ R_{i,t,\text{day}} > 0, \\ 0 & \text{otherwise.} \end{cases} \tag{11}$$

where $ToW_t^{\text{alternative},(i)}$ is an indicator variable equal to one when stock i experiences a positive intraday reversal on day t. We aggregate this measure across stocks to identify the corresponding n- and q-days, replacing the original tug-of-war measure in Equation (2). We then repeat the Fama and MacBeth (1973) estimation for the newly identified n-days.

The results are consistent with the prediction. The untabulated analysis yields a strongly positive SML slope on these newly defined n-days ($\bar{\lambda}^{(n\text{-day})} = 0.14$, t-stat $= 6.37$). More importantly, we do not reject the slope restriction ($\bar{\lambda}^{(n\text{-day})} - (\bar{R}_{m,t}^{(n\text{-day})} - \bar{R}_{f,t}^{(n\text{-day})}) = -0.014$, t-stat $= -0.78$). Thus, unlike the original n-days characterized by negative intraday reversals, these days do not exhibit a statistically significant departure from the CAPM.

These results provide additional evidence that the CAPM failure documented on the original n-days is not simply a consequence of any intraday reversal. Rather, it is specifically the negative intraday reversal, following a positive overnight return, that is associated with the over-correction mechanism. The absence of a similar pattern following a positive intraday reversal is consistent with the proposed economic mechanism and helps rule out a generic reversal-based explanation for our findings.

V Alternative Explanations

In this section, we investigate whether our findings can be explained by existing mechanisms that have been shown to generate conditional failures of the CAPM. We first consider whether our results on n-days reflect the absence of fundamental news, such as macroeconomic announcements (Savor and Wilson, 2014) or earnings news (Chan and Marsh, 2022). We then examine explanations related to intraday return patterns (Hendershott et al., 2020),

speculative beta (Hong and Sraer, 2016), and beta compression (Andrei et al., 2023). For each explanation, we outline its implications for our analysis and provide evidence that challenges its ability to account for our findings.

A Fundamental News

Savor and Wilson (2014) document that the CAPM holds on macroeconomic announcement days but is rejected on non-macroeconomic news days. Similarly, Chan and Marsh (2022) show that the CAPM holds on LEADs but fails on non-LEADs. If n-days simply captured non-fundamental-news days, our findings would mechanically replicate those documented in prior studies.

To examine this possibility, we first analyze the overlap between n-days and days without fundamental news. The last four rows of Panel D of Table 1 show that 50.09% of non-macroeconomic announcement days and 53.71% of non-LEADs are classified as q-days. Thus, the absence of fundamental news is not sufficient to generate an n-day classification, casting doubt on the possibility that our results are driven by a simple distinction between news and non-news days.

We further repeat our asset pricing tests after excluding days associated with macroeconomic announcements. Panel B of Table 4 shows that the slope restriction is rejected on n-days ($\bar{\lambda}^{(n\text{-day})} - (\bar{R}_{m,t}^{(n\text{-day})} - \bar{R}_{f,t}^{(n\text{-day})}) = -0.12$, t-stat $= -5.20$), but not on q-days ($\bar{\lambda}^{(q\text{-day})} - (\bar{R}_{m,t}^{(q\text{-day})} - \bar{R}_{f,t}^{(q\text{-day})}) = 0.01$, t-stat $= 0.64$). A similar pattern emerges when we exclude LEADs. Panel B of Table 5 shows that the slope restriction is rejected on n-days, ($\bar{\lambda}^{(n\text{-day})} - (\bar{R}_{m,t}^{(n\text{-day})} - \bar{R}_{f,t}^{(n\text{-day})}) = -0.12$, t-stat $= -4.14$), while it is not rejected on q-days ($\bar{\lambda}^{(q\text{-day})} - (\bar{R}_{m,t}^{(q\text{-day})} - \bar{R}_{f,t}^{(q\text{-day})}) = 0.01$, t-stat $= 0.24$).

Taken together, these findings indicate that our results are distinct from those documented by Savor and Wilson (2014) and Chan and Marsh (2022). If our findings merely reflected a reclassification of non-fundamental-news days, we would expect the CAPM failure to emerge broadly among non-news days, rather than being concentrated specifically among n-days.

B Time of Day Effects

Hendershott et al. (2020) argue that the CAPM holds during the overnight trading period but fails during the intraday trading period. One possible explanation for our findings is therefore that the CAPM failure on n-days simply reflects the greater importance of intraday returns on these days. Several pieces of evidence, however, are difficult to reconcile with this interpretation.

First, Panel A of Table 3 shows that the slope restriction is rejected not only during the intraday period ($\bar{\lambda}^{(n\text{-day})} - (\bar{R}_{m,t}^{(n\text{-day})} - \bar{R}_{f,t}^{(n\text{-day})}) = -0.18, t\text{-stat} = -9.06$) but also during the overnight period ($\bar{\lambda}^{(n\text{-day})} - (\bar{R}_{m,t}^{(n\text{-day})} - \bar{R}_{f,t}^{(n\text{-day})}) = 0.06, t\text{-stat} = 5.28$) on n-days.

Second, if the CAPM failure on n-days were simply a manifestation of the intraday effect documented by Hendershott et al. (2020), we would expect the CAPM to fail for intraday returns more generally. However, Panel B of Table 3 shows that the intraday slope restriction is rejected on n-days ($\bar{\lambda}^{(n\text{-day})} - (\bar{R}_{m,t}^{(n\text{-day})} - \bar{R}_{f,t}^{(n\text{-day})}) = -0.18, t\text{-stat} = -9.06$) but not on q-days ($\bar{\lambda}^{(q\text{-day})} - (\bar{R}_{m,t}^{(q\text{-day})} - \bar{R}_{f,t}^{(q\text{-day})}) = 0.01, t\text{-stat} = 0.32$). Thus, the intraday CAPM failure is not a general feature of intraday returns; rather, it emerges only on n-days identified by our aggregate tug-of-war measure.

Overall, these results suggest that the time-of-day effect documented by Hendershott

et al. (2020) is insufficient to explain our findings. The CAPM failure does not arise simply because returns are measured during the intraday period; rather, it is concentrated on days characterized as n-days.

C Speculative Beta

We examine whether the speculative beta mechanism proposed by Hong and Sraer (2016) can explain our findings. In their model, investors disagree about the common factor driving firms' cash flows. Because high-beta stocks have greater exposure to this factor, beta amplifies the impact of disagreement about macroeconomic outcomes on investors' valuations. Consequently, disagreement generates greater dispersion in the valuations of high-beta stocks, making them more susceptible to overpricing. In equilibrium, short-sale constraints prevent pessimistic investors from fully expressing their views, so high-beta stocks are predominantly held by optimistic investors. Although arbitrageurs attempt to correct this mispricing, their limited risk-bearing capacity prevents them from eliminating it completely. As aggregate disagreement increases, speculative overpricing becomes more pronounced among high-beta stocks, flattening the SML and, when disagreement is sufficiently high, potentially generating a negative relation between beta and excess returns.

If this mechanism drives our results, we would expect the slope restriction to be systematically rejected for both n- and q-days during periods of high disagreement. To test this, we measure firm-level disagreement as the standard deviation of analysts' long-term earnings growth forecasts (Diether et al., 2002). We then construct an aggregate disagreement measure as the beta-weighted cross-sectional average of firm-level disagreement (Hong and Sraer, 2016). We classify a month as high disagreement if aggregate disagreement in

the previous month exceeds its full-sample median. All remaining months are classified as low disagreement months. Panel A of Table 7 shows that the slope restriction is rejected on n-days ($\bar{\lambda}^{(n\text{-day})} - (\bar{R}_{m,t}^{(n\text{-day})} - \bar{R}_{f,t}^{(n\text{-day})}) = -0.09$, t-stat $= -4.04$) but not on q-days ($\bar{\lambda}^{(q\text{-day})} - (\bar{R}_{m,t}^{(q\text{-day})} - \bar{R}_{f,t}^{(q\text{-day})}) = -0.01$, t-stat $= -0.47$), suggesting that the rejection of the slope restriction is not uniform across n- and q-days that form the high disagreement sample.

We next focus on n-days and examine the performance of the CAPM separately during periods of high and low disagreement. Panel B of the same table documents that the rejection of the slope restriction is economically stronger on n-days during periods of low disagreement ($\bar{\lambda}^{(n\text{-day})} - (\bar{R}_{m,t}^{(n\text{-day})} - \bar{R}_{f,t}^{(n\text{-day})}) = -0.15$, t-stat $= -4.12$), than during periods of high disagreement. Thus, the severity of the CAPM failure does not increase with investor disagreement. This evidence is inconsistent with the speculative beta hypothesis, which predicts that disagreement-induced beta mispricing should intensify during periods of elevated disagreement.

D Beta Compression

Andrei et al. (2023) show that the information gap between the empiricist and investors can make the CAPM appear to fail empirically, even when it holds from investors' perspectives. Specifically, they find that when investors disagree about expected returns, beta estimates are biased. On disagreement days, the econometrician overestimates the betas of high-beta assets and underestimates those of low-beta securities. As a result, the SML rotates clockwise. When the disagreement is resolved, beta compression occurs: the beta estimate of the high-beta asset decreases and that of the low-beta asset increases. They validate their results by comparing beta estimates on FOMC press conference days and other days. One

might wonder if this mechanism could explain the difference between the performance of the CAPM on n- and q- days.

There are several reasons to be skeptical about this possibility. First, Panel D of Table 1 shows that n- and q- days are equally likely to occur on macro and non-macro days. Second, we compare the averages of the conditional beta estimates obtained separately on n- and q-days, following the approach of Andrei et al. (2023). Figure 3 shows that the estimates are very similar, and we find no evidence of beta compression on q-days, casting doubt on the beta compression mechanism. If the beta compression argument were valid, the 45-degree line would not provide an accurate description of the data.

VI Conclusion

This paper provides new evidence on time-variation in the performance of the CAPM. We introduce a novel indicator capturing market-level tug-of-war and use it to classify trading days into two distinct types: q-days and n-days. Our analysis reveals a fundamental difference in the behavior of the SML across these two types of days. On q-days, the SML slopes upward, consistent with the CAPM, whereas it slopes downward on n-days. More importantly, the slope restriction, which requires the estimated market risk premium to equal the average market realized excess return, is not rejected on q-days but is rejected on n-days. These findings are robust to alternative research design choices.

Taken together, our findings show that the CAPM performs well on q-days but fails on n-days. Importantly, our results suggest that the scope of the CAPM failure is narrower than previously documented. Prior studies have documented CAPM failures for returns mea-

sured over various periods, including non-macroeconomic announcement days, non-earnings announcement days, and intraday intervals (Savor and Wilson, 2014; Chan and Marsh, 2022; Hendershott et al., 2020). Our findings show that these failures are not pervasive across time but are concentrated in a subset of days characterized by market-level tug-of-war dynamics. Consequently, the CAPM successfully describes the cross-section of returns over a much larger fraction of trading days than previously recognized. We further show that the over-correction hypothesis of Akbas et al. (2022) provides an explanation for the failure of the model on n-days.

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Figure 1: Security Market Line on n- and q-Days

This figure shows the relationship between the average daily excess return of the test assets, computed over either noisy (n) or quiet (q) days, and their average beta estimates. The asset menu consists of 10 (value-weighted) beta-sorted portfolios. The vertical axis shows the average excess returns of each asset for the subsamples of n- or q-days, represented by the blue and red colors, respectively. At the end of each month, we use a trailing window of 252 days to estimate the conditional beta of each portfolio. We then calculate the sample mean of the conditional beta estimates and report it on the horizontal axis. This beta estimate is used for both n- and q- days. For each type of day, the line visually represents the empirical security market line obtained after estimating the cross-sectional regression of the average portfolio excess returns calculated over the corresponding day-type, i.e., n- or q-day, on an intercept and the average portfolio beta estimates. The sample period runs from July 1992 to December 2022.

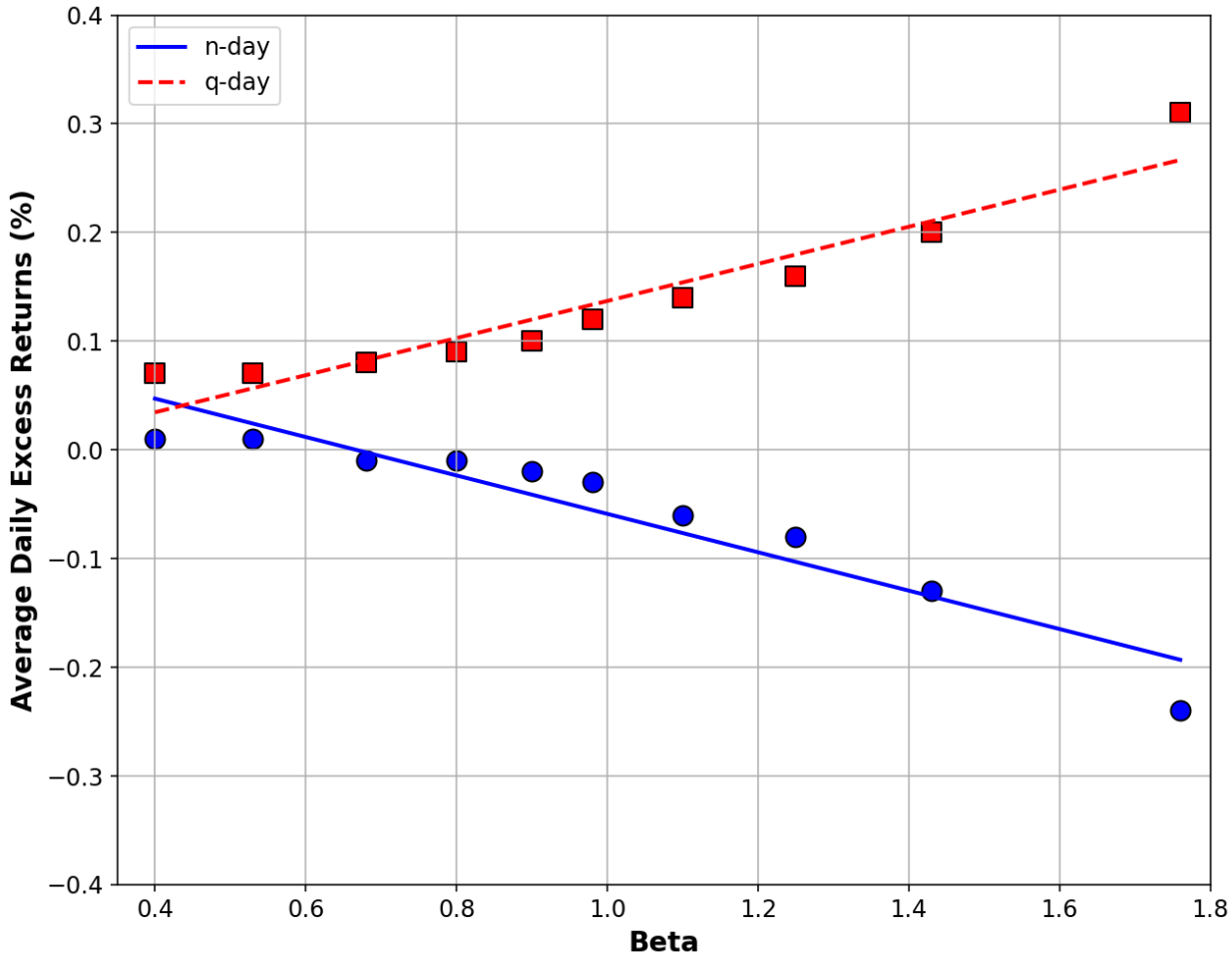


Figure 2: Share of n-Days Over Time

This figure shows the fraction of trading days, expressed in percentage points, in each calendar year classified as n-days. The sample period is from July 1992 to December 2022.

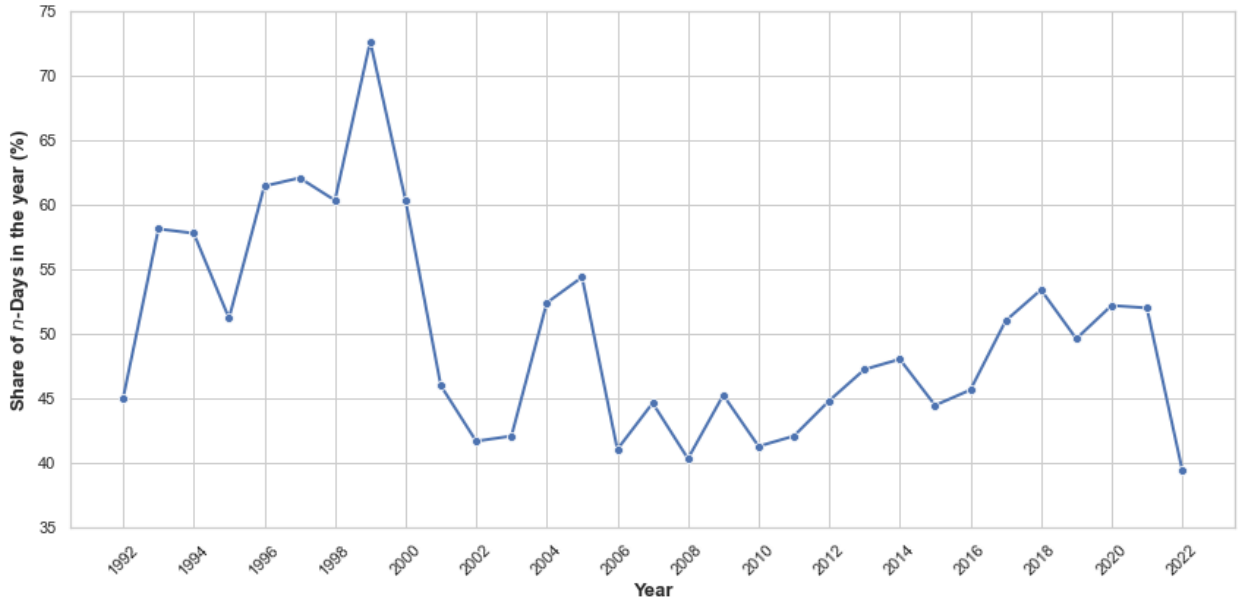


Figure 3: Beta estimates on n- and q-Days

This figure shows the relationship between the beta estimates of each test asset based on the sample of n- and q-days, respectively. The vertical axis shows the average conditional beta estimates linked to q-days, while the horizontal axis shows the average conditional beta estimates across n-days. The asset menu consists of 10 value-weighted and beta-sorted portfolios. The sample period is from July 1992 to December 2022.

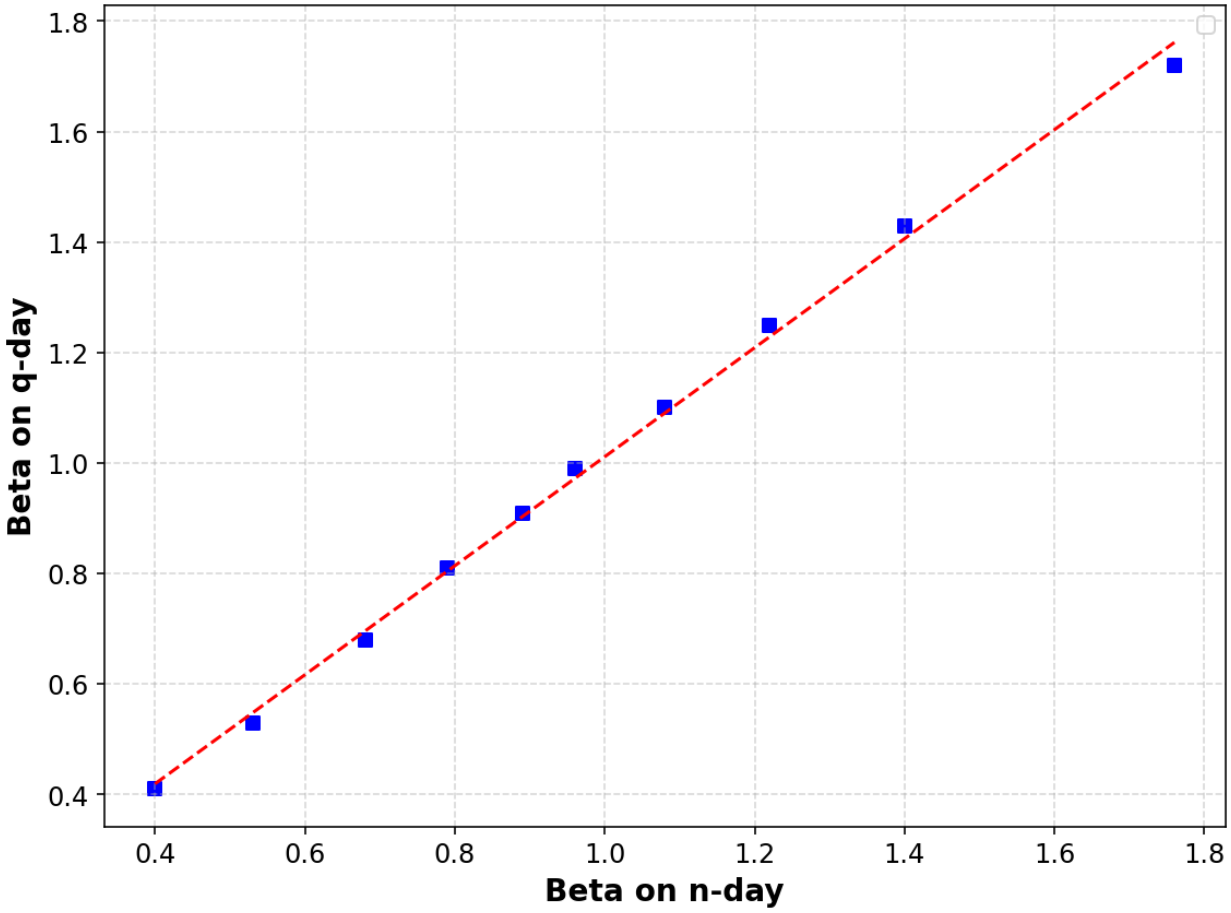


Table 1: Summary Statistics

This table presents key statistics. Panel A characterizes the aggregate tug-of-war variable (ToW^m). *Mean* and *Std Dev* denote the mean and standard deviation, respectively. *AR(1)*, *AR(2)*, and *AR(3)* are the first, second, and third order autocorrelations of ToW^m, respectively. *Min*, *Max*, and *Median* are the minimum, maximum, and median of ToW^m, respectively. Panel B reports the proportion of trading days [name in column] that are identified as an n-day. Panel C documents the proportion of trading days in the month [name in column] that are classified as n-days. Finally, Panel D summarizes the means of the market excess return, the TED spread, the VIX index, and the risk aversion index (Bekaert et al., 2022) on n- and q-days. The next two rows report the proportions of macroeconomic (federal funds rate, producer price index, and unemployment rate; see Savor and Wilson (2014)) and non-macroeconomic news days, respectively, classified under each day type. The final two rows show the corresponding proportions for leading earnings announcement (LEAD) and non-LEAD news days. Following Chan and Marsh (2022), LEAD days are defined as those falling between Tuesday and Thursday of the first week of the earnings reporting quarter, provided that at least 50 S&P 500 firms announce earnings during that week.

Panel A. ToW^m

Mean	*Std Dev*	*AR(1)*	*AR(2)*	*AR(3)*	*Min*	*Max*	*Median*
0.28	0.12	−0.06	−0.02	−0.02	0.03	0.86	0.28

Panel B. Share of n-days on Each Day of the Week

Monday	*Tuesday*	*Wednesday*	*Thursday*	*Friday*
52.42%	50.60%	49.02%	48.58%	49.54%

Panel C. Share of n-days in Each Month of the Year

Jan	*Feb*	*Mar*	*Apr*	*May*	*Jun*	*Jul*	*Aug*	*Sep*	*Oct*	*Nov*	*Dec*
51.72%	51.48%	52.35%	52.02%	47.87%	53.21%	49.16%	48.83%	48.49%	50.59%	46.29%	48.17%

Panel D. Economic Variables

	$n-day$	$q-day$
Market Excess Return (%)	−0.05	0.12
TED (%)	0.45	0.45
VIX (%)	19.30	19.97
Risk Aversion	2.98	3.11
% Macro Days	50.63%	49.37%
% Non-Macro Days	49.91%	50.09%
% LEAD Days	47.92%	52.08%
% Non-LEAD Days	46.29%	53.71%

Table 2: Security Market Line: 10 (Value-Weighted) Beta-sorted Portfolios

This table summarizes the results of the asset pricing tests applied to the 10 (value-weighted) beta-sorted portfolios. Panel A focuses on the Fama and MacBeth (1973) methodology. At the end of each month, we use a trailing window of daily excess returns observed over the past 252 days to estimate the (post-ranking) conditional beta of each test asset. For each trading day in the next month, we estimate a cross-sectional regression of the excess returns of the test assets on a constant and the estimated betas. We repeat these steps each month and obtain the time-series of the estimated coefficients. $\bar{\alpha}$ is the mean of the time-series of the intercept estimates. $\bar{\lambda}$ is the mean of the time-series of the slope estimates. *Adj* R^2 is the mean adjusted R^2. Each entry in the column refers to the sample of estimates recorded over days [name in row]. $\bar{R}_m - \bar{R}_f$ denotes the mean market excess return over the days [name in row]. $\bar{\lambda} - (\bar{R}_m - \bar{R}_f)$ shows the difference between the estimated market risk premium and the average realized market excess return over the days [name in row]. Panel B shows the results of the pooled regression of portfolio excess returns on a constant and the explanatory variable(s) (see Equations (8)–(9)). All returns are expressed in percentage points per day. Parentheses in Panel A show Newey and West (1987) t-statistics with a Bartlett truncated at 10 lags; parentheses in Panel B show the t-statistics based on standard errors clustered at the daily level. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively.

Panel A. Fama and MacBeth (1973) Estimation

Type of Day	$\bar{\alpha}$	$\bar{\lambda}$	*Adj* R^2	$\bar{R}_m - \bar{R}_f$	$\bar{\lambda} - (\bar{R}_m - \bar{R}_f)$
n-day	0.06	−0.13	40.46%	−0.05	−0.08
	(3.15)	(−4.23)		(−3.23)	(−3.60)
q-day	−0.00	0.15	45.66%	0.12	0.02
	(−0.25)	(5.15)		(5.39)	(1.22)

Panel B. Pooled Estimation

α	λ	ψ	ϕ	*Adj* R^2
0.05	−0.01			0.00%
(3.25)	(−0.33)			
−0.00	0.14	0.10	−0.30	0.65%
(−0.01)	(4.48)	(3.58)	(−7.65)	

Table 3: Dissecting the SML: Intraday versus Overnight

This table summarizes the results of the overnight and intraday SML, based on the 45 portfolios, on n-days. At the end of each month, we use a trailing window of daily excess returns observed over the past 252 days to estimate the (post-ranking) conditional beta of each test asset. For each n-day in the next month, we estimate a cross-sectional regression of the [name in row] excess returns of the test assets on a constant and the estimated betas. Following Lou et al. (2024), we compute intraday and overnight excess returns by subtracting the appropriate fraction of the daily risk-free rate from raw returns, 6.5/24 for the intraday trading window and 17.5/24 for the overnight period. We repeat these steps each month and obtain the time-series of the estimated coefficients. $\bar{\alpha}$ is the mean of the time-series of the intercept estimates. $\bar{\lambda}$ is the mean of the time-series of the slope estimates. *Adj* R^2 is the mean adjusted R^2. $\bar{R}_m - \bar{R}_f$ denotes the mean market excess return over the days [name in row]. $\bar{\lambda} - (\bar{R}_m - \bar{R}_f)$ shows the difference between the estimated market risk premium and the average realized market excess return over the days [name in row]. Panel A reports the corresponding results for n-days during intraday and overnight periods, whereas Panel B reports the results for the intraday period linked to n- and q-days separately. All returns are expressed in percentage points per day. In parentheses, we report the Newey and West (1987) t-statistics based on a Bartlett truncated at 10 lags. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively.

Panel A. Dissecting the n-day SML: Intraday versus Overnight

Period	$\bar{\alpha}$	$\bar{\lambda}$	*Adj* R^2	$\bar{R}_m - \bar{R}_f$	$\bar{\lambda} - (\bar{R}_m - \bar{R}_f)$
Overnight	−0.04	0.36	24.05%	0.31	0.06
	(−4.20)	(19.10)		(38.38)	(5.28)
Intraday	0.16	−0.55	23.58%	−0.37	−0.18
	(8.48)	(−20.45)		(−27.19)	(−9.06)

Panel B. Intraday SML: n-days versus q-days

Period	$\bar{\alpha}$	$\bar{\lambda}$	*Adj* R^2	$\bar{R}_m - \bar{R}_f$	$\bar{\lambda} - (\bar{R}_m - \bar{R}_f)$
Intraday (n-day)	0.16	−0.55	23.58%	−0.37	−0.18
	(8.48)	(−20.45)		(−27.19)	(−9.06)
Intraday (q-day)	0.02	0.38	24.62%	0.37	0.01
	(0.91)	(14.45)		(21.98)	(0.32)

Table 4: Macroeconomic News

This table reports the results of the asset pricing tests for the 45 test portfolios on macroeconomic announcement days and other days. Macroeconomic announcement days are those associated with announcements of the (i) producer price index, (ii) unemployment rate, and (iii) federal funds rate. We refer to all remaining trading days as other days. For each type of day, we identify the n- and q-days and repeat our main asset pricing tests. At the end of each month, we estimate the post-ranking conditional beta of each test asset using a trailing window of 252 daily excess returns. For each trading day in the subsequent month, we estimate a cross-sectional regression of test asset excess returns on a constant and the estimated betas. Repeating this procedure each month yields a time series of estimated coefficients. $\bar{\alpha}$ is the mean of the time-series of the intercept estimates, $\bar{\lambda}$ is the mean of the time-series of the slope estimates, and *Adj.* R^2 denotes the average adjusted R^2 across the cross-sectional regressions. Each entry in the column refers to the sample of estimates recorded over days [name in row]. $\bar{R}_m - \bar{R}_f$ denotes the mean market excess return over the days [name in row]. $\bar{\lambda} - (\bar{R}_m - \bar{R}_f)$ shows the difference between the estimated market risk premium and the average realized market excess return over the days [name in row]. Panel A reports the results for n-days occurring on macroeconomic announcement days and for n-days occurring on other days. Panel B reports the results for both n- and q-days within the sample of other days. All returns are expressed in percentage points per day. In parentheses, we report the Newey and West (1987) t-statistics based on a Bartlett truncated at 10 lags. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively.

Panel A. n-days: Macroeconomic versus Other Days

Type of Day	$\bar{\alpha}$	$\bar{\lambda}$	*Adj* R^2	$\bar{R}_m - \bar{R}_f$	$\bar{\lambda} - (\bar{R}_m - \bar{R}_f)$
Macroeconomic Days	0.05	−0.08	23.72%	−0.03	−0.07
	(1.02)	(−0.98)		(−0.29)	(−1.37)
Other Days	0.09	−0.18	21.63%	−0.06	−0.12
	(3.96)	(−5.74)		(−3.35)	(−5.20)

Panel B. Other Days: n-days versus q-days

Type of Day	$\bar{\alpha}$	$\bar{\lambda}$	*Adj* R^2	$\bar{R}_m - \bar{R}_f$	$\bar{\lambda} - (\bar{R}_m - \bar{R}_f)$
n-day	0.09	−0.18	21.63%	−0.06	−0.12
	(3.96)	(−5.74)		(−3.35)	(−5.20)
q-day	0.01	0.12	26.12%	0.11	0.01
	(0.54)	(3.72)		(4.31)	(0.64)

Table 5: Leading Earnings Announcement

This table reports the results of the asset pricing tests for the 45 test portfolios on leading earnings announcement days (LEADs) and other days. The LEADs are the days between Tuesday and Thursday of the first week of the (earnings) reporting quarter that has a minimum of 50 (S&P 500) announcers. We refer to all remaining trading days as other days. For each type of day, we identify the n- and q-days and repeat our main asset pricing tests. At the end of each month, we estimate the post-ranking conditional beta of each test asset using a trailing window of 252 daily excess returns. For each trading day in the subsequent month, we estimate a cross-sectional regression of test asset excess returns on a constant and the estimated betas. Repeating this procedure each month yields a time series of estimated coefficients. $\bar{\alpha}$ is the mean of the time-series of the intercept estimates, $\bar{\lambda}$ is the mean of the time-series of the slope estimates, and *Adj.* R^2 denotes the average adjusted R^2 across the cross-sectional regressions. Each entry in the column refers to the sample of estimates recorded over days [name in row]. $\bar{R}_m - \bar{R}_f$ denotes the mean market excess return over the days [name in row]. $\bar{\lambda} - (\bar{R}_m - \bar{R}_f)$ shows the difference between the estimated market risk premium and the average realized market excess return over the days [name in row]. Panel A reports the results for n-days occurring on LEADs and for n-days occurring on other days. Panel B reports the results for both n- and q-days within the sample of other days. All returns are expressed in percentage points per day. In parentheses, we report the Newey and West (1987) t-statistics based on a Bartlett truncated at 10 lags. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively.

Panel A. n-days: LEADs versus Other Days

Type of Day	$\bar{\alpha}$	$\bar{\lambda}$	*Adj* R^2	$\bar{R}_m - \bar{R}_f$	$\bar{\lambda} - (\bar{R}_m - \bar{R}_f)$
LEADs	−0.24	0.12	19.60%	−0.07	0.19
	(−1.78)	(0.69)		(−0.80)	(1.42)
Other Days	0.08	−0.15	23.24%	−0.03	−0.12
	(3.04)	(−3.77)		(−1.61)	(−4.14)

Panel B. Other Days: n-days versus q-days

Type of Day	$\bar{\alpha}$	$\bar{\lambda}$	*Adj* R^2	$\bar{R}_m - \bar{R}_f$	$\bar{\lambda} - (\bar{R}_m - \bar{R}_f)$
n-day	0.08	−0.15	23.24%	−0.03	−0.12
	(3.04)	(−3.77)		(−1.61)	(−4.14)
q-day	0.02	0.09	26.43%	0.08	0.01
	(0.84)	(2.39)		(2.85)	(0.24)

Table 6: Sentiment

This table presents the results of the asset pricing tests applied to the 45 portfolios over different sentiment periods. We identify a trading month as optimistic (pessimistic) if the Baker and Wurgler (2006) sentiment index measured at the end of the previous month is positive (negative). For each sentiment period, i.e., pessimistic or optimistic, we identify the n- and q-days and repeat our main asset pricing tests. At the end of each month, we use a trailing window of daily excess returns observed over the past 252 days to estimate the (post-ranking) conditional beta of each test asset. For each trading day in the next month, we estimate a cross-sectional regression of the excess returns of the test assets on a constant and the estimated betas. We repeat these steps each month and obtain the time-series of the estimated coefficients. $\bar{\alpha}$ is the mean of the time-series of the intercept estimates. $\bar{\lambda}$ is the mean of the time-series of the slope estimates. *Adj* R^2 is the mean adjusted R^2. Each entry in the column refers to the sample of estimates recorded over days [name in row]. $\bar{R}_m - \bar{R}_f$ denotes the mean market excess return over the days [name in row]. $\bar{\lambda} - (\bar{R}_m - \bar{R}_f)$ shows the difference between the estimated market risk premium and the average realized market excess return over the days [name in row]. Panel A reports the results for n-days during periods of pessimistic and optimistic sentiment, whereas Panel B reports the corresponding results for periods of optimistic sentiment. All returns are expressed in percentage points per day. In parentheses, we report the Newey and West (1987) t-statistics based on a Bartlett truncated at 10 lags. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively.

Panel A. n-days: Pessimistic versus Optimistic Periods

Period	$\bar{\alpha}$	$\bar{\lambda}$	*Adj* R^2	$\bar{R}_m - \bar{R}_f$	$\bar{\lambda} - (\bar{R}_m - \bar{R}_f)$
Pessimistic	0.07	−0.13	21.68%	−0.02	−0.11
	(2.46)	(−3.02)		(−0.72)	(−3.54)
Optimistic	0.10	−0.26	22.69%	−0.14	−0.13
	(4.83)	(−7.84)		(−6.18)	(−5.77)

Panel B. Optimistic Periods: n-days versus q-days

Type of Day	$\bar{\alpha}$	$\bar{\lambda}$	*Adj* R^2	$\bar{R}_m - \bar{R}_f$	$\bar{\lambda} - (\bar{R}_m - \bar{R}_f)$
n-day	0.10	−0.26	22.69%	−0.14	−0.13
	(4.83)	(−7.84)		(−6.18)	(−5.77)
q-day	−0.01	0.23	25.60%	0.19	0.04
	(−0.24)	(6.52)		(6.42)	(1.55)

Table 7: Aggregate Disagreement

This table reports the results of the asset pricing tests for the 45 test portfolios across different aggregate disagreement periods. Following Diether et al. (2002), we measure firm-level disagreement as the standard deviation of analysts' long-term earnings growth forecasts. Aggregate disagreement is then computed as the beta-weighted cross-sectional average of firm-level disagreement (Hong and Sraer, 2016). We classify a month as a high disagreement month if aggregate disagreement in the previous month exceeds its full-sample median; all remaining months are classified as low disagreement months. At the end of each month, we estimate the post-ranking conditional beta of each test asset using a trailing window of 252 daily excess returns. For each trading day in the subsequent month, we estimate a cross-sectional regression of test asset excess returns on a constant and the estimated betas. Repeating this procedure each month yields a time series of estimated coefficients. $\bar{\alpha}$ is the mean of the time-series of the intercept estimates, $\bar{\lambda}$ is the mean of the time-series of the slope estimates, and *Adj.* R^2 denotes the average adjusted R^2 across the cross-sectional regressions. Each column reports estimates for the sample indicated by the corresponding row. $\bar{R}_m - \bar{R}_f$ denotes the mean market excess return over the days [name in row]. $\bar{\lambda} - (\bar{R}_m - \bar{R}_f)$ shows the difference between the estimated market risk premium and the average realized market excess return over the days [name in row]. Panel A reports the results for both n- and q-days during periods of high aggregate disagreement. Panel B reports the results for n-days during periods of high and low aggregate disagreement. All returns are expressed in percentage points per day. In parentheses, we report the Newey and West (1987) t-statistics based on a Bartlett truncated at 10 lags. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively.

Panel A. High Disagreement Periods: n-days versus q-days

Type of Day	$\bar{\alpha}$	$\bar{\lambda}$	*Adj* R^2	$\bar{R}_m - \bar{R}_f$	$\bar{\lambda} - (\bar{R}_m - \bar{R}_f)$
n-day	0.07	−0.26	23.49%	−0.18	−0.09
	(2.99)	(−7.72)		(−7.60)	(−4.04)
q-day	0.04	0.23	28.52%	0.24	−0.01
	(1.78)	(6.10)		(7.64)	(−0.47)

Panel B. n-days: Low versus High Disagreement Periods

Period	$\bar{\alpha}$	$\bar{\lambda}$	*Adj* R^2	$\bar{R}_m - \bar{R}_f$	$\bar{\lambda} - (\bar{R}_m - \bar{R}_f)$
Low Disagreement	0.10	−0.07	20.80%	0.08	−0.15
	(3.18)	(−1.54)		(3.25)	(−4.12)
High Disagreement	0.07	−0.26	23.49%	−0.18	−0.09
	(2.99)	(−7.72)		(−7.60)	(−4.04)

Market-Level Tug of War and Asset Pricing

Online Appendix

JEL classification: G11, G12, G41

Keywords: Capital Asset Pricing Model, Over-correction, Security Market Line, Tug-of-War

Figure A1: Bootstrap Results

This figure shows the histogram of the estimated market risk premium based on 1,000 pseudo-samples of n-days characterized by a positive average realized market excess return. The bootstrap aims to create a pseudo-sample of the same length as our sample of n-days. To achieve this goal, we draw (with replacement) from the sample of n-days and only retain the pseudo-sample with a positive average realized market excess return. We then run our Fama and MacBeth (1973) estimation and compute the estimated risk premium. We repeat the above steps 1,000 times and report the distribution of the estimated risk premium.

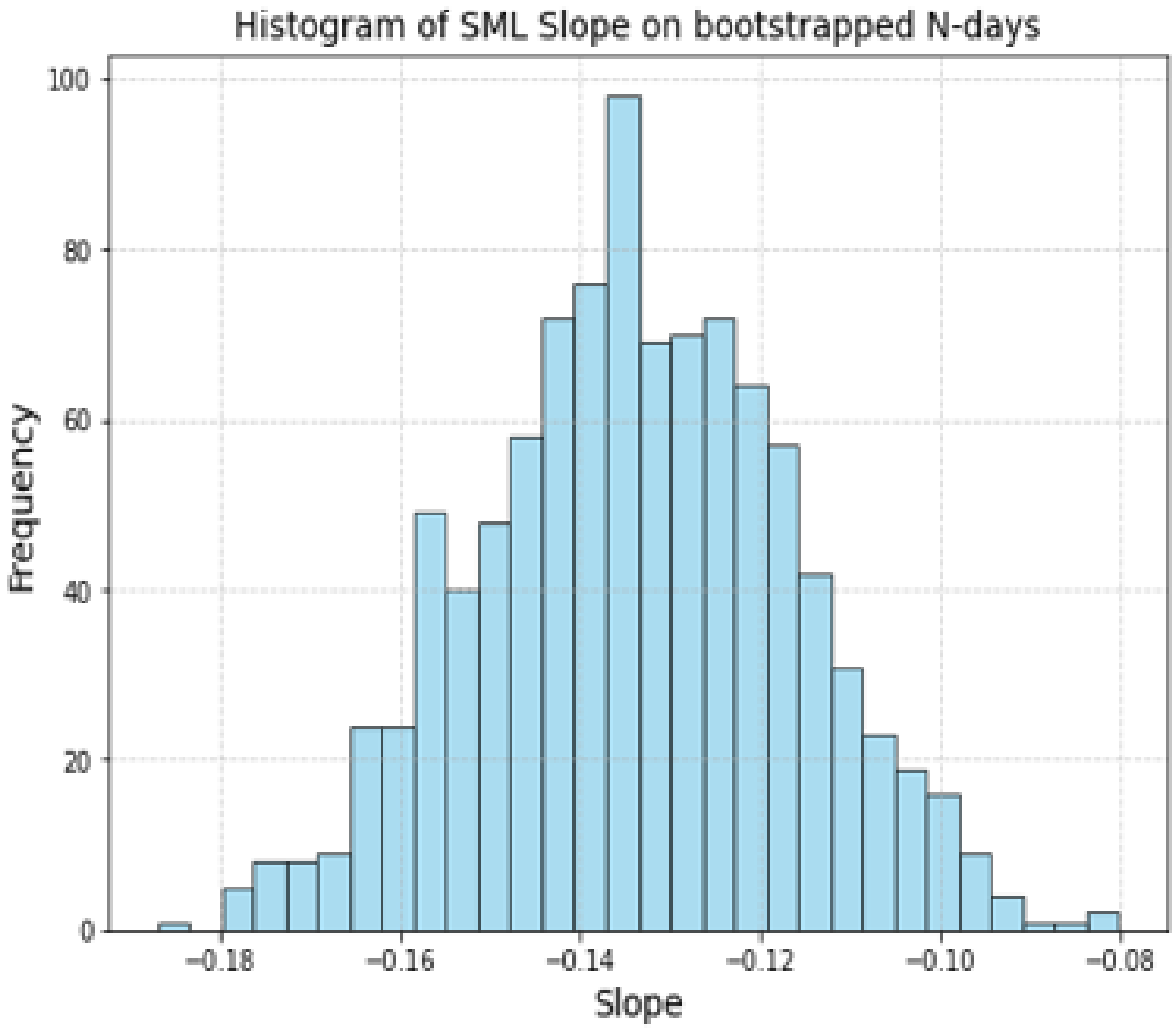


Table A1: Security Market Line (Full Sample Analysis)

This table summarizes the results of the Fama and MacBeth (1973) asset pricing tests using the test assets described in [name in row]. At the end of each month, we use a trailing window of daily excess returns observed over the past 252 days to estimate the (post-ranking) conditional beta of each test asset. For each trading day in the next month, we estimate a cross-sectional regression of the excess returns of the test assets on a constant and the estimated betas. We repeat this procedure each month to obtain the time-series of the estimated coefficients. $\bar{\alpha}$ is the mean of the time-series of the intercept estimates. $\bar{\lambda}$ is the mean of the time-series of the slope estimates. *Adj* R^2 is the mean adjusted R^2. All returns are expressed in percentage points per day. Parentheses contain Newey and West (1987) t-statistics computed using Bartlett kernel standard errors with a truncation lag of 10. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively.

Test Assets	$\bar{\alpha}$	$\bar{\lambda}$	*Adj* R^2
10 Value-Weighted Beta-Sorted	0.03 (2.28)	0.01 (0.56)	43.08%
10 Equal-Weighted Beta-Sorted	0.13 (9.61)	−0.05 (−2.57)	50.21%
Individual Equities	0.07 (7.48)	0.00 (0.29)	2.18%
Individual Equities (Controlling for Book-to-Market, Size, and Momentum)	0.25 (8.19)	0.02 (1.40)	3.30%
45 Portfolios	0.04 (3.15)	−0.01 (−0.39)	24.29%

Table A2: Security Market Line: 10 (Equal-Weighted) Beta-sorted Portfolios

This table summarizes the results of the asset pricing tests applied to the 10 (equal-weighted) beta-sorted portfolios. Panel A focuses on the Fama and MacBeth (1973) methodology. At the end of each month, we use a trailing window of daily excess returns observed over the past 252 days to estimate the (post-ranking) conditional beta of each test asset. For each trading day in the next month, we estimate a cross-sectional regression of the excess returns of the test assets on a constant and the estimated betas. We repeat these steps each month and obtain the time-series of the estimated coefficients. $\bar{\alpha}$ is the mean of the time-series of the intercept estimates. $\bar{\lambda}$ is the mean of the time-series of the slope estimates. *Adj* R^2 is the mean adjusted R^2. Each entry in the column refers to the sample of estimates recorded over days [name in row]. $\bar{R}_m - \bar{R}_f$ denotes the mean market excess return over the days [name in row]. $\bar{\lambda} - (\bar{R}_m - \bar{R}_f)$ shows the difference between the estimated market risk premium and the average realized market excess return over the days [name in row]. Panel B shows the results of the pooled regression of portfolio excess returns on a constant and the explanatory variable(s) (see Equations (8)–(9)). All returns are expressed in percentage points per day. Parentheses in Panel A show Newey and West (1987) t-statistics with a Bartlett truncated at 10 lags; parentheses in Panel B show the t-statistics based on standard errors clustered at the daily level. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively.

Panel A. Fama and MacBeth (1973) Estimation

Type of Day	$\bar{\alpha}$	$\bar{\lambda}$	*Adj* R^2	$\bar{R}_m - \bar{R}_f$	$\bar{\lambda} - (\bar{R}_m - \bar{R}_f)$
n-day	0.18	−0.25	47.22%	−0.05	−0.20
	(10.10)	(−8.17)		(−3.23)	(−9.34)
q-day	0.07	0.15	53.16%	0.12	0.03
	(3.93)	(5.70)		(5.39)	(1.65)

Panel B. Pooled Estimation

α	λ	ψ	ϕ	*Adj* R^2
0.14	−0.06			0.03%
(9.39)	(−3.10)			
0.12	0.10	0.06	−0.34	1.09%
(4.97)	(3.28)	(2.14)	(−8.86)	

Table A3: Security Market Line: Individual Equities

This table summarizes the results of the asset pricing tests applied to the individual equities. Panel A focuses on the Fama and MacBeth (1973) methodology. At the end of each month, we use a trailing window of daily excess returns observed over the past 252 days to estimate the conditional beta of each individual equity. For each trading day in the next month, we estimate a cross-sectional regression of the excess returns of the test assets on a constant and the estimated betas. We repeat these steps each month and obtain the time-series of the estimated coefficients. $\bar{\alpha}$ is the mean of the time-series of the intercept estimates. $\bar{\lambda}$ is the mean of the time-series of the slope estimates. *Adj* R^2 is the mean adjusted R^2. Each entry in the column refers to the sample of estimates recorded over days [name in row]. $\bar{R}_m - \bar{R}_f$ denotes the mean market excess return over the days [name in row]. $\bar{\lambda} - (\bar{R}_m - \bar{R}_f)$ shows the difference between the estimated market risk premium and the average realized market excess return over the days [name in row]. Panel B controls for established characteristics, such as the market capitalization (*Size*), the Book-to-Market (BTM), and the momentum (MOM). Panel C shows the results of the pooled regression of individual equity excess returns on a constant and the explanatory variable(s) (see Equations (8)–(9)). All returns are expressed in percentage points per day. Parentheses in Panels A and B show Newey and West (1987) t-statistics with a Bartlett truncated at 10 lags; parentheses in Panel C show the t-statistics based on standard errors clustered at the daily level. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively.

Panel A. Fama and MacBeth (1973)

Type of Day	$\bar{\alpha}$	$\bar{\lambda}$	*Adj* R^2	$\bar{R}_m - \bar{R}_f$	$\bar{\lambda} - (\bar{R}_m - \bar{R}_f)$
n-day	0.10	−0.16	1.15%	−0.06	−0.11
	(7.42)	(−9.23)		(−3.62)	(−8.57)
q-day	0.12	0.10	1.77%	0.13	−0.02
	(6.59)	(5.68)		(5.64)	(−1.78)

Panel B. Fama and MacBeth (1973) with Controls

Type of Day	$\bar{\alpha}$	$\bar{\lambda}$	*Size*	*BTM*	*MOM*	*Adj* R^2	$\bar{R}_m - \bar{R}_f$	$\bar{\lambda} - (\bar{R}_m - \bar{R}_f)$
n-day	0.38	−0.13	−0.02	0.03	−0.00	2.11%	−0.06	−0.08
	(6.20)	(−8.01)	(−4.98)	(6.31)	(−2.47)		(−3.62)	(−5.80)
q-day	0.39	0.11	−0.02	−0.01	−0.00	2.76%	0.13	−0.01
	(7.20)	(6.55)	(−6.23)	(−2.71)	(−0.12)		(5.64)	(−0.86)

Panel C. Pooled Regression

α	λ	ψ	ϕ	*Size*	*BTM*	*MOM*	*Adj* R^2
0.10	−0.03						0.03%
(6.88)	(−2.77)						
0.15	0.08	−0.03	−0.28				0.19%
(6.68)	(4.33)	(−1.38)	(−11.71)				
1.54	0.11	−0.04	−0.27	−0.11	0.02	−0.00	0.22%
(16.60)	(6.26)	(−1.57)	(−11.66)	(−14.94)	(2.15)	(−3.25)	

Table A4: Security Market Line: 45 Portfolios

This table summarizes the results of the asset pricing tests applied to the 45 portfolios. Panel A focuses on the Fama and MacBeth (1973) methodology. At the end of each month, we use a trailing window of daily excess returns observed over the past 252 days to estimate the (post-ranking) conditional beta of each test asset. For each trading day in the next month, we estimate a cross-sectional regression of the excess returns of the test assets on a constant and the estimated betas. We repeat these steps each month and obtain the time-series of the estimated coefficients. $\bar{\alpha}$ is the mean of the time-series of the intercept estimates. $\bar{\lambda}$ is the mean of the time-series of the slope estimates. *Adj* R^2 is the mean adjusted R^2. Each entry in the column refers to the sample of estimates recorded over days [name in row]. $\bar{R}_m - \bar{R}_f$ denotes the mean market excess return over the days [name in row]. $\bar{\lambda} - (\bar{R}_m - \bar{R}_f)$ shows the difference between the estimated market risk premium and the average realized market excess return over the days [name in row]. Panel B shows the results of the pooled regression of portfolio excess returns on a constant and the explanatory variable(s) (see Equations (8)–(9)). All returns are expressed in percentage points per day. Parentheses in Panel A show Newey and West (1987) t-statistics with a Bartlett truncated at 10 lags; parentheses in Panel B show the t-statistics based on standard errors clustered at the daily level. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively.

Panel A. Fama and MacBeth (1973) Estimation

Type of Day	$\bar{\alpha}$	$\bar{\lambda}$	*Adj* R^2	$\bar{R}_m - \bar{R}_f$	$\bar{\lambda} - (\bar{R}_m - \bar{R}_f)$
n-day	0.08	−0.17	22.25%	−0.05	−0.12
	(4.16)	(−5.71)		(−3.23)	(−5.46)
q-day	−0.00	0.15	26.30%	0.12	0.03
	(−0.08)	(5.06)		(5.39)	(1.38)

Panel B. Pooled Estimation

α	λ	ψ	ϕ	*Adj* R^2
0.04	−0.01			0.00%
(2.36)	(−0.30)			
0.06	0.09	−0.01	−0.21	0.68%
(2.06)	(2.83)	(−0.37)	(−5.18)	

Table A5: Alternative Threshold: 0.25

This table summarizes the results of the asset pricing tests applied to the 45 portfolios. We identify a trading day as an n-day if the aggregate tug-of-war on that day is above 0.25. All other days are q-days. Panel A focuses on the Fama and MacBeth (1973) methodology. At the end of each month, we use a trailing window of daily excess returns observed over the past 252 days to estimate the (post-ranking) conditional beta of each test asset. For each trading day in the next month, we estimate a cross-sectional regression of the excess returns of the test assets on a constant and the estimated betas. We repeat these steps each month and obtain the time-series of the estimated coefficients. $\bar{\alpha}$ is the mean of the time-series of the intercept estimates. $\bar{\lambda}$ is the mean of the time-series of the slope estimates. *Adj* R^2 is the mean adjusted R^2. Each entry in the column refers to the sample of estimates recorded over days [name in row]. $\bar{R}_m - \bar{R}_f$ denotes the mean market excess return over the days [name in row]. $\bar{\lambda} - (\bar{R}_m - \bar{R}_f)$ shows the difference between the estimated market risk premium and the average realized market excess return over the days [name in row]. Panel B shows the results of the pooled regression of portfolio excess returns on a constant and the explanatory variable(s) (see Equation (9)). All returns are expressed in percentage points per day. Parentheses in Panel A show Newey and West (1987) t-statistics with a Bartlett truncated at 10 lags; parentheses in Panel B show the t-statistics based on standard errors clustered at the daily level. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively.

Panel A. Fama and MacBeth (1973) Estimation

Type of Day	$\bar{\alpha}$	$\bar{\lambda}$	*Adj* R^2	$\bar{R}_m - \bar{R}_f$	$\bar{\lambda} - (\bar{R}_m - \bar{R}_f)$
n-day	0.07	−0.09	22.43%	0.01	−0.10
	(4.22)	(−3.45)		(0.43)	(−5.21)
q-day	−0.01	0.12	27.18%	0.08	0.04
	(−0.45)	(3.21)		(2.99)	(1.61)

Panel B. Pooled Estimation

α	λ	ψ	ϕ	*Adj* R^2
0.03	0.07	0.03	−0.14	0.18%
(0.94)	(1.98)	(0.65)	(−3.23)	

Table A6: Alternative Threshold: 70th Percentile

This table summarizes the results of the asset pricing tests applied to the 45 portfolios. We identify a trading day as an n-day if the aggregate tug-of-war on that day is above the 70th percentile of its distribution. All other days are q-days. Panel A focuses on the Fama and MacBeth (1973) methodology. At the end of each month, we use a trailing window of daily excess returns observed over the past 252 days to estimate the (post-ranking) conditional beta of each test asset. For each trading day in the next month, we estimate a cross-sectional regression of the excess returns of the test assets on a constant and the estimated betas. We repeat these steps each month and obtain the time-series of the estimated coefficients. $\bar{\alpha}$ is the mean of the time-series of the intercept estimates. $\bar{\lambda}$ is the mean of the time-series of the slope estimates. *Adj* R^2 is the mean adjusted R^2. Each entry in the column refers to the sample of estimates recorded over days [name in row]. $\bar{R}_m - \bar{R}_f$ denotes the mean market excess return over the days [name in row]. $\bar{\lambda} - (\bar{R}_m - \bar{R}_f)$ shows the difference between the estimated market risk premium and the average realized market excess return over the days [name in row]. Panel B shows the results of the pooled regression of portfolio excess returns on a constant and the explanatory variable(s) (see Equation (9)). All returns are expressed in percentage points per day. Parentheses in Panel A show Newey and West (1987) t-statistics with a Bartlett truncated at 10 lags; parentheses in Panel B show the t-statistics based on standard errors clustered at the daily level. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively.

Panel A. Fama and MacBeth (1973) Estimation

Type of Day	$\bar{\alpha}$	$\bar{\lambda}$	*Adj* R^2	$\bar{R}_m - \bar{R}_f$	$\bar{\lambda} - (\bar{R}_m - \bar{R}_f)$
n-day	0.10	−0.31	22.31%	−0.16	−0.16
	(3.64)	(−7.76)		(−7.54)	(−5.29)
q-day	0.01	0.13	25.19%	0.12	0.01
	(0.83)	(5.59)		(6.76)	(0.43)

Panel B. Pooled Estimation

α	λ	ψ	ϕ	*Adj* R^2
0.07	0.07	−0.10	−0.24	1.30%
(3.25)	(2.79)	(−2.79)	(−5.74)	

Table A7: Alternative Aggregate Tug-of-War

This table summarizes the results of the asset pricing tests applied to the 45 portfolios. We compute the aggregate tug-of-war directly at the market level (See Equation (10)). We then use this alternative measure to identify n- and q-days. Panel A focuses on the Fama and MacBeth (1973) methodology. At the end of each month, we use a trailing window of daily excess returns observed over the past 252 days to estimate the (post-ranking) conditional beta of each test asset. For each trading day in the next month, we estimate a cross-sectional regression of the excess returns of the test assets on a constant and the estimated betas. We repeat these steps each month and obtain the time-series of the estimated coefficients. $\bar{\alpha}$ is the mean of the time-series of the intercept estimates. $\bar{\lambda}$ is the mean of the time-series of the slope estimates. *Adj* R^2 is the mean adjusted R^2. Each entry in the column refers to the sample of estimates recorded over days [name in row]. $\bar{R}_m - \bar{R}_f$ denotes the mean market excess return over the days [name in row]. $\bar{\lambda} - (\bar{R}_m - \bar{R}_f)$ shows the difference between the estimated market risk premium and the average realized market excess return over the days [name in row]. Panel B shows the results of the pooled regression of portfolio excess returns on a constant and the explanatory variable(s) (see Equation (9)). All returns are expressed in percentage points per day. Parentheses in Panel A show Newey and West (1987) t-statistics with a Bartlett truncated at 10 lags; parentheses in Panel B show the t-statistics based on standard errors clustered at the daily level. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively.

Panel A. Fama and MacBeth (1973) Estimation

Type of Day	$\bar{\alpha}$	$\bar{\lambda}$	*Adj* R^2	$\bar{R}_m - \bar{R}_f$	$\bar{\lambda} - (\bar{R}_m - \bar{R}_f)$
n-day	0.10	−0.42	20.98%	−0.31	−0.11
	(3.23)	(−10.75)		(−18.32)	(−3.49)
q-day	0.02	0.14	25.46%	0.16	−0.02
	(1.32)	(6.29)		(8.73)	(−1.27)

Panel B. Pooled Estimation

α	λ	ψ	ϕ	*Adj* R^2
0.05	0.11	−0.01	−0.46	2.24%
(2.26)	(4.26)	(−0.19)	(−11.17)	

Table A8: Security Market Line: Individual Equities (Weighted Least Squares)

This table summarizes the results of the asset pricing tests applied to the individual equities. Panel A focuses on the weighted least squares estimation of Asparouhova et al. (2010). At the end of each month, we use a trailing window of daily excess returns observed over the past 252 days to estimate the (post-ranking) conditional beta of each test asset. For each trading day in the next month, we estimate a weighted cross-sectional regression of the excess returns of the test assets on a constant and the estimated betas. In estimating this regression, we use the sum of 1 and the lagged return of each asset as weights. We repeat these steps each month and obtain the time-series of the estimated coefficients. $\bar{\alpha}$ is the mean of the time-series of the intercept estimates. $\bar{\lambda}$ is the mean of the time-series of the slope estimates. *Adj* R^2 is the mean adjusted R^2. Each entry in the column refers to the sample of estimates recorded over days [name in row]. $\bar{R}_m - \bar{R}_f$ denotes the mean market excess return over the days [name in row]. $\bar{\lambda} - (\bar{R}_m - \bar{R}_f)$ shows the difference between the estimated market risk premium and the average realized market excess return over the days [name in row]. Panel B controls for established characteristics, such as the market capitalization (*Size*), the Book-to-Market (*BTM*) and the momentum (*MOM*). All returns are expressed in percentage points per day. Parentheses in Panel A show Newey and West (1987) t-statistics with a Bartlett truncated at 10 lags; parentheses in Panel B show the t-statistics based on standard errors clustered at the daily level. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively.

Panel A. Fama and MacBeth (1973) Estimation

Type of Day	$\bar{\alpha}$	$\bar{\lambda}$	*Adj* R^2	$\bar{R}_m - \bar{R}_f$	$\bar{\lambda} - (\bar{R}_m - \bar{R}_f)$
n-day	0.08	−0.15	1.14%	−0.06	−0.10
	(6.06)	(−8.70)		(−3.62)	(−7.71)
q-day	0.10	0.11	1.78%	0.13	−0.02
	(5.45)	(6.19)		(5.64)	(−1.13)

Panel B. Fama and MacBeth (1973) with Controls

Type of Day	$\bar{\alpha}$	$\bar{\lambda}$	*Size*	*BTM*	*MOM*	*Adj* R^2	$\bar{R}_m - \bar{R}_f$	$\bar{\lambda}-(\bar{R}_m-\bar{R}_f)$
n-day	0.29	−0.13	−0.02	0.03	−0.00	2.10%	−0.06	−0.08
	(4.65)	(−7.84)	(−3.57)	(6.15)	(−1.72)		(−3.62)	(−5.48)
q-day	0.28	0.12	−0.02	−0.01	0.00	2.75%	0.13	−0.01
	(5.22)	(6.77)	(−4.28)	(−2.30)	(0.70)		(5.64)	(−0.60)